1. Excellence

1.1 Objectives and ambition

The main objective of the RECREATE project is to develop a set of innovative technologies aimed at exploiting the potential of End-of-Life (EoL) complex composite waste (mainly carbon fiber reinforced composites CFRC, and glass fiber reinforced composites GFRC) as a feedstock for profitable reuse of parts and materials in the manufacturing industry. The EU composites market size is actually foreseen to grow steadily with an annual growth rate of 7.5% in the next few years. Legislation is progressively banning landfilling in many countries and the amount of composite waste to be recycled and reused will be huge in the near future, with an estimated >80'000 ton in 2025.[1] On the other side, there is an increasing market demand of high performance fiber materials (especially carbon) at affordable costs, in many sectors like automotive/transportation and in general for the lightweight design field. It is therefore crucial that new recycling technologies alternative to the more consolidated mechanical grinding and pyrolysis are identified, so as to allow the recovery and reuse of materials and components without compromise to downcycling, and with convincing environmental sustainability and economic profitability. In the light of these considerations, the ambition of RECREATE project is:

- to develop and validate in relevant environment (TRL6) novel reuse strategies for current generation, large EoL composite parts (including complex multimaterial composites) based on smart recognition and inspection for sorting (Laser Induced Breakdown Spectroscopy - LIBS), high precision dismantling (laser-shock) and repair, T-assisted reshaping, design for disassembly based on reversible joints, AI-assisted decision support systems;
- to develop and validate in relevant environment (TRL6) innovative physico-chemical upcycling technologies (catalyst-assisted green solvolysis, electrofragmentation) allowing simultaneous recovery of high quality, integer, clean fibers and of an organic resin fraction reusable as coating material, at the very end of the multiple reuse processes of parts;
- to demonstrate at TRL6 the use of smart and green reversible thermoset resins as enabling materials for the realization of the next generation of fiber-reinforced composites (FRCs) with easier repairability and enhanced reusability, facilitating the transition towards recyclable-by-design composite materials and structures.

Moreover, RECREATE will address another key objective, that is the development of a set of new digital tools for:

- the quantitative evaluation of the environmental and economic performance of the proposed technologies (LCA/LCC) as well as their circularity assessment;
- the co-design of innovative digital learning resources, including the realization of MOOCS, serious games and digital twins of some specialty technologies developed in the project, with easy adoption and high replicability.

The objectives and the ambition of RECREATE are fully compliant with the general requirements of the Horizon Europe - Digital, Industry and Space 2021 Work Programme, and with the specific requirements of the call Horizon-CL4-2021-Resilience-01-01, as shown schematically in the following **Table 1**.

Table 1. Project adherence to the work-programme and call: objectives, expected results and means of verifications.

Specific challenge of Work Programme. Call HORIZON-CL4-2021-Resilience-01-01	Adherence to the call	Objectives and expected results	Means of verification (KPI of the technology)
Propose innovative dismantling and sorting systems enabling reuse and functional recycling of complex composite materials.	RECREATE will adopt and bring to TRL6 innovative, laser-based technologies for dismantling, recognition and sorting, including: - laser-shock dismantling - LIBS	**Laser-shock dismantling**	
		Demonstration at 0.5 m^2 scale on one interface, including process and integrity control (TRL6).	Post-treatment material integrity (< 10% loss in tensile modulus; < 20% loss in flexural modulus).
		Laser-induced breakdown spectroscopy - LIBS	
		CFRC/GFRC composite recognition in waste flows (TRL6).	Flow rate recognition of 25 kg/h.
Develop and integrate novel solutions for higher reuse of the whole products and components.	RECREATE will develop and adopt different technological approaches for the remanufacturing and reuse of large composite components: - T-assisted reshaping; - laser-based machining; - design for disassembly based on reversible joints; - non-destructive	**T-assisted reshaping**	
		flatten curved composite parts with the following characteristics: - max. GFRC curvature radius: 400 mm; - typ. GFRC EoL part thickness: 2-5 mm; - content GF: ≥ 30 wt.%.	Deflection on flat panels ≤ 10 mm on 1200 mm length.
		Laser-based machining	

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	disassembly/reassembly of complex composite parts. Cutting, joining and remanufacturing pathways will be driven by AI-based decision support systems.	Surface manufacturing of FRCs at ply scale.	Post-treatment material integrity (< 10% loss in tensile modulus; < 20% loss in flexural modulus).
		Design for disassembly based on reversible joints	
		Efficient T-triggered separation of dissimilar materials; compatibility with current processing equipment and tools.	Residual strength < 1 MPa in bonded assembly after < 10 min at triggering T; dismantling efficiency > 99%.
		Non-destructive dis-/reassembly of complex FRC parts	
		Efficient process *via* a new process chain demonstrated on wind rotor blades (T5.3).	Disassembly time < 6 h per blade; reassembly time < 4 h per blade; 0 % damage to dismantled components.
Develop novel, safe, environment friendly and commercially attractive methods of recycling a wide range of composite materials and reuse of secondary raw materials	RECREATE will develop novel approaches for recycling complex composite materials and parts based on: - catalyst-assisted green solvolysis - electrofragmentation RECREATE will also develop new strategies to enable effective use of secondary raw materials through the incorporation of new reversible green resins, adhesives and joints in new generation FRCs. Commercial attractiveness as well as environmental friendliness of these technologies will be demonstrated at TRL6 *via* dedicated environmental and socio-economic sustainability assessment (WP6).	**Catalyst-assisted green solvolysis**	
		New green solvolysis process (ambient pressure, moderate T, non-toxic/biobased solvents, polymer-tailored catalyst selection) demonstrated in a relevant operating environment (TRL6).	Purity of reclaimed fibers > 90%, evaluated by means of thermogravimetric analysis (TGA); recovered organic fraction > 50% vs. matrix weight.
		Electrofragmentation	
		Process optimization *via* DoE tailored for different materials and EoL waste streams (TRL6); process modelling, digital simulation and data acquisition for process control and optimization; process automation.	Purity of reclaimed fibers > 90%, evaluated by means of thermogravimetric analysis (TGA).
Demonstrate at pilot level the feasibility of reuse and/or recycle approaches of composites and its secondary raw materials.	RECREATE will develop: 4 physical demonstrators based on the reuse of GFRC/CFRC EoL parts after reshaping, repair and reversible assembly technologies; 5 physical demonstrators based on the use of innovative upcycling technologies for recovery and reuse of reinforcing fibers and polymer resins.	**Demonstrators: reuse of GFRC/CFRC EoL parts**	
		Target industrial sectors: - transport; - wind energy; - aviation	Please refer to Table 2 for detailed KPIs.
		Demonstrators: fibers and resin recovery and reuse	
		Target industrial sectors: - aviation; - sport; - industrial coatings	Please refer to Table 2 for detailed KPIs.
Develop tools that will enable to demonstrate the circularity and the environmental benefits of the solutions tested.	WP6 in RECREATE will be dedicated to development and implementation of digital tools (IoT, big data, data analytics) to assess circularity of the proposed technologies. Such tools will be integrated into a freeware to be applied to the demonstrators and after project end will be available to stakeholders under license.	**Digital tools for circularity assessment & acceleration**	
		A commercial software integrating digital tools for assessing and accelerating circularity of FRC reuse and recycling technologies.	Application of the software by all industrial demo-case owners and end-users. Distribution of at least 2 software licenses per year during the project.
Consider the co-design of learning	WP7 in RECREATE will be	**Co-design of learning resources and digital tools**	

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resources together with local and regional educational organizations for current and future generations of employees.	dedicated to the development of innovative learning resources and tools. WP7 will be led by GRIFO and supported by METID unit @POLIMI, expert in design and testing tools for learning innovation and faculty development. The tools will be co-designed with the main local education stakeholders, and will be tested and replicated through the network of academic contacts of POLIMI and other partners involved in the project.	The expected results include: innovative learning resources (e.g., e-tutorials and e-learning modules); a simulation game providing hands-on experience on the reuse of complex composite materials, to be fed with real data; realization of educational digital twins	Evaluation tests included in the learning resources; 3 experts of the macro-areas related to the topics involved; 2 focus groups organized to test the prototypes leading to digital twin first release; reference standard ISO 9241-11:2018 for validation; completion rate over 75%; number of errors below 25%; system usability scale > 75%.

The main progress beyond the state-of-the-art for all **key technologies** in RECREATE is summarized below.

Temperature-assisted reshaping. Thermoset polymers like unsaturated polyesters (U-PEs) are known to be not thermo-processable due to the irreversibility of the chemical crosslinking reaction. However recent research carried out at ICAM has found that it is possible to develop a cost-effective process to reuse large EoL parts made of PEs/vinylester GFRCs with slightly curved geometry, avoiding grinding and crushing (Fig.1).[2] The new technology development is based on reshaping large parts of polyester composites with simple geometry using pressure and temperature at maximum resin deformation point. PEs-based cured resins typically exhibit a glass transition temperature (Tg) ~100°C, which allows the reshaping process without polymer degradation. This reshaping process is affected by several factors: crosslinking density of the resin; chemical structure; type/geometry of composite part and reinforcement (% GFs, orientation, etc.); composite thickness. A deep knowledge of stress relaxation behaviour of materials and associated process variables is essential to fully exploit the potential of this new technology. The reshaping process will be brought up to TRL6 via a multi-step multi-technology approach: T-assisted deformation of the material to the required geometry (through calendering, compression molding); holding at the same temperature for sufficiently long time to release most of the residual stresses without thermal degradation; controlled cooling below Tg while maintaining the final planar shape. A further advancement will be the synergistic development of the reshaping technology along with the development and demonstration of new resin systems for future-generation FRC materials (WP4).

Fig. 1 Preliminary (TRL3) results on T-assisted reshaping of 400mm radius GFRCs to flatter parts using a twin roll.

Reversible adhesives. The traditional methods for joining plastics are mechanical fastening, adhesive joining, and welding. Mechanical joining *via* fasteners, such as screws, enables EoL disassembly. Unfortunately, its disadvantages include increasing product weight. Furthermore, dismantling fastened products can be expensive and time consuming and often needs to be carried out manually. Adhesive joining has many benefits including process speed, low weight, low cost, design flexibility, improved stiffness of the joint, ability to damp noise and vibrations, uniform distribution of stresses over the assembled areas, and possibility to join dissimilar materials, as targeted in RECREATE. The most common strategies to obtain adhesive debonding on command are represented by hot-melt adhesives (not suitable for structural applications), modified resins (limited scalability), formulations with debonding fillers like thermally expandable particles. RESCOLL's patented technology, called INDAR Inside®, is based on custom formulations of adhesives and primers including blowing agents.[3] Such proprietary materials allow high mechanical strength (15-20 MPa on Al) at room T and easy dismantling (residual strength <1 MPa) after heating at a specific T in the 100-200 °C range, depending on the use case and the corresponding service temperatures. In RECREATE, RESCOLL will develop a new range of dedicated new INDAR adhesives (or primers) based on on-the-shelf formulations. To avoid any anticipated triggering, premature disassembly and loss of debonding capability, it will be essential to keep a T margin (at least

Fig. 2 Heating patches applied for debonding nautical rear-foils from the engine bonded with INDAR technology (left) and debonded foil (right).

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50 °C) between maximum service T and targeted debonding T. The debonding process will be developed so that no residual adhesive remains on one or both surfaces after dismantling. To date, RESCOLL's technology has been successfully implemented in various applications at lower TRL (Fig.2 is an example of its use in the ship industry for composite foil bonding).

Design for disassembly. Successful design for dissasembly is based on three different disciplines: selection and use of materials; geometric design of components, structures or products; selection and use of joints and fasteners (Fig.3). Multimaterial design, with the use of different materials (e.g., metals and GFRCs) in a single component, has been a major research focus in the field of lightweight design in recent years. However, in the framework of circular economy and the highest possible recycling and reuse rates, components made of one material are to be preferred.[4] The current challenge is to combine these two conflicting requirements and to enable optimal (hybrid) components and structures that are easy to recycle. Detachable joints, especially adhesive joints, can contribute to the solution. In this way, different materials can be joined together and separated again after the EoL.

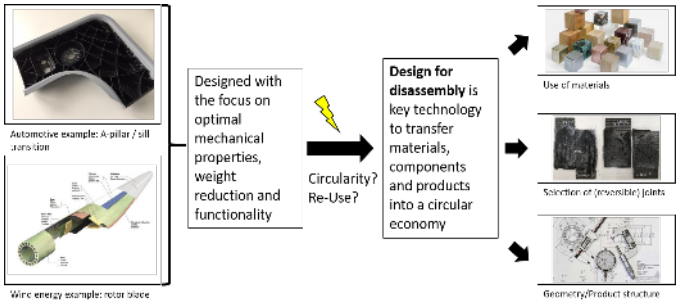

Fig. 3 Design for disassembly approach followed in RECREATE.

Laser-based high precision dismantling and machining. Laser dismantling consists in focusing a high power laser (ns, GW/cm^2) on the surface of the multilayer structure to create high pressure plasma (GPa) (Fig.4). A shock wave is induced in the material. By playing with the laser propagation length and time, a tensile forces can be produced at any location inside the material with high precision, including at the interface of FRC laminates or multilayer assemblies. The extension of such process to dismantling consists in applying a laser shock over the entire surface of the targeted jointed surface to delaminate the material. No contact and local mechanical tensile stresses ensure material integrity for part re-use or recycling. In addition to be employed as active dismantling tool, high power lasers can also be exploited a source of energy allowing material machining. By controlling the parameters of the laser (duration, wavelength) and of the process (gas assistance, laser beam moving) it will be possible to repair surfaces, assemble materials or cut them in non-contact mode.[5,6] Moreover laser technology can allow highly robotized processes. Effort must also focus on the integrity of the target surfaces for flawless reuse of materials for second-gen structures. In this framework, activities in RECREATE will be directed towards the development at higher TRL of hybrid processes associating delamination by laser shock and laser-based FRC machining. In both cases, modeling of the processes *via* digital twins will allow optimization and online control of the process with additional sensors.

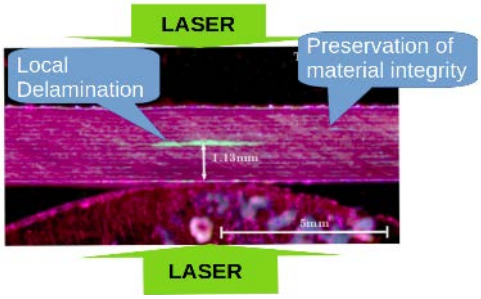

Fig. 4 Representative image of localized laser-based dismantling and delamination of a complex FRC structure.

Laser induced Breakdown Spectroscopy (LIBS). Industrial sectors as automotive, aviation, nautical etc. are not using yet well-established integrated processes for the recycling of EoL complex GFRC/CFRC composites due to the insufficient and/or ineffective existing recognition solution capacities. Current technologies for the characterization and analysis of plastics in general are mainly based on optical spectroscopies.[7] Vis-NIR spectroscopy and hyper-spectra imaging (HIS) are the most commonly used so far, but these techniques cannot deal for ex. with the detection of black materials. Nevertheless, emerging technologies have aroused interest from recycling companies as they have the potential to bridge this gap (MIR-HSI, THz imaging, LIBS etc.). While MIR-HSI system requires a high investment and THz imaging is limited by a low specificity and the available bandwidths, LIBS shows a good compatibility with demanding industrial environments (both batch and inline).[8-10] However, the similar elemental composition of most polymers results in high similarity of their LIBS spectra, which makes their discrimination challenging. To address this issue, RECREATE will go beyond the State-of-Art (SoA) by adopting optical spectroscopy techniques combined with powerful artificial intelligence (AI) algorithms and advanced chemometric models. The current limit of detection, efficiency and inline integrability will be increased. IRIS has already demonstrated the ability to use some of these techniques developing in-line prototypes (e.g., SWIR-HSI

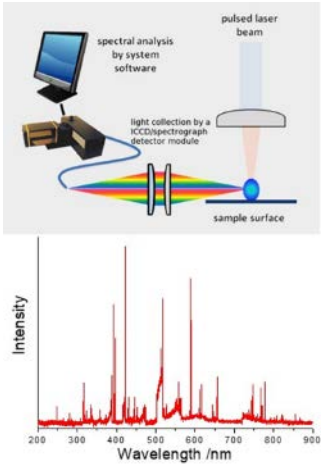

Fig. 5 (top) LIBS setup: a pulsed laser is focused on a sample; (bottom) example of a LIBS spectrum, with peaks corresponding to specific atomic lines.

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for identifying the different layers in laminated plastic films as well as LIBS for identifying black plastics in the H2020 Multicycle project). Starting from TRL4, IRIS will aim to further the LIBS technology for inline industrial applications up to TRL6 (Fig.5).

Catalyst-assisted green solvolysis. Solvolysis is a well know technology (classified at TRL 6) for the recovery of high quality fibers from EoL composites.[11] The main disadvantage is the generation of a large amount of organic waste to be disposed of, mainly by combustion, with generation of large amount of CO_2. Moreover, solvolysis in supercritical or near supercritical fluids (high T and pressure) is needed in order to break-up linkages more recalcitrant to depolymerization. Attempts have been made recently to develop solvolysis in green, biobased solvents like limonene.[12] More interestingly, it has been found that catalyst-assisted solvolysis is able to recycle FRCs efficiently at an ordinary pressure and mild T, therefore minimizing costs and fiber damage.[13] Research at POLIMI has recently demonstrated at lab level (TRL3/4) the possibility to carry out an efficient solvolysis of CFRCs in optimized conditions to recover simultaneously both high quality fibers and an organic fraction reusable as coating.[14] The approach proposed in RECREATE is thus to develop in a relevant operating environment (TRL6) a **new green solvolysis process at ambient pressure and moderate T**, with a careful selection of catalysts (depending on polymer resin type) and solvents (non-toxic solvents, possibly biobased). The new process will synergistically close the loop, enabling recovery of undamaged clean fibers and use of the liquid organic fraction as component in coating formulations. **Solvent selection will be driven by industrial (coating) and environmental requirement. To maximize sustainability of the process, recycling of solvent will be pursued during coating formulation and preparation relying on solvent recirculation/recycling approaches well-consolidated in the coating industry.**

Electrofragmentation. One of the innovative recycling technologies valorized during RECREATE is the high voltage electro-dynamic fragmentation (from now on HVF).[15] HVF is an innovative technology, which exploits a fast-growing voltage to overcome the dielectric breakdown threshold of the product under recycling and to generate multiple spark channels within the product itself. HVF has been tested with polyester-based GFRCs.[16] Experimental results highlight that the process is particularly suitable for the treatment of multiphase complex materials as FRCs. The combination of different effects as the creation of the spark channel nearby the phase border (because of higher electric field variation) and the shockwave driven disaggregation of the components magnified by different acoustic impedances between the phases result in an output composed by cleaner fibres, longer fibre length distribution, higher percentage of fibres at mean fibre length and lower retained resin content than composites recycled mechanically. HVF technology could be explored to recover not only reinforcement fibers but also the resin. Despite the HVF technology is quite affirmed today in sectors such as mining and minerals purifications, the technology is only available at lab scale for recycling at low TRL4 and several researches has to be carried on different fronts. The main advancements addressed in RECREATE will include: possibility to recover also the resin component; process parameter optimization *via* DoE tailored for different materials and EoL waste streams, in view of the use of HVF to treat entire products and not just parts (TRL6); process modelling, digital simulation and data acquisition leading to process control and optimization; hardware modification, including process automation.

Vitrimers and reversible green resins. Recent progresses in polymer chemistry offer innovative pathways that may help to improve the EoL and waste management of composites by *predictive material design*. In particular, the incorporation of *dynamic crosslinking bonds* in the macromolecular network of polymer resins has been recently introduced as powerful chemical tool to equip thermosets with new thermally-triggered functionalities potentially enabling their repairing, welding, reprocessing and reuse in suitable and controllable moderate temperature ranges.[17] Accordingly, RECREATE project will exploit this concept to develop new smart polymeric systems to be used as reversible matrix polymers in a new generation of FRCs, more compliant to the principles of circular economy and successfully exhibiting multiple repair, reuse and de/remanufacturing capabilities when heated at specific processing temperatures, without affecting their usability and functionality. The dynamic covalent chemistry approaches investigated in RECREATE will be based on *covalent adaptable networks* (CAN) including reversible diene-dienophile Diels-Alder (D-A) dissociative polymers and associative macromolecular vitrimers (Fig.6).[18,19]

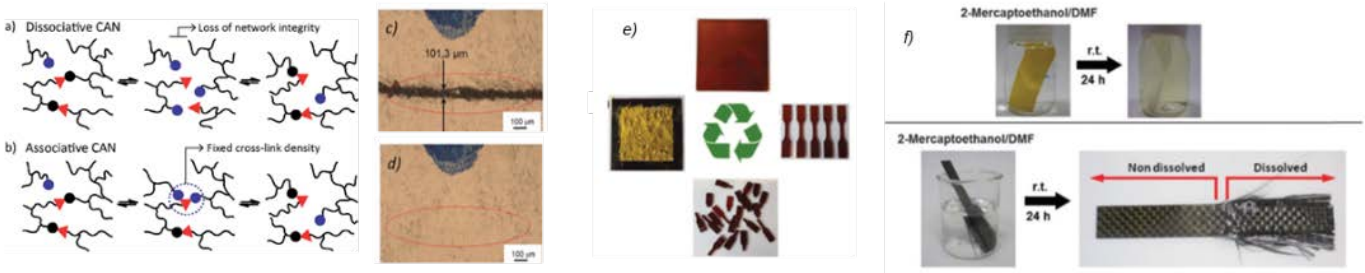

Fig. 6 (a) Schematic representation of dissociative (D-A) and (b) associative (e.g., vitrimers) CANs; (c-d) OM images of the thermally-induced scratch healing process of a D-A polymer; (e) dynamic thermosetting epoxy matrix recycling; (f) complete solvolysis of dynamic epoxy matrix and recovery of CFs from CFRC.

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While being thermosets at working T, such dynamic materials may undergo straightforward solvolysis in specific ranges of mild temperatures (130-160 °C), enabling direct recovery of clean undamaged fibres as well as of reusable resin matrix material. Additionally, the dynamic character of such systems will enable multiple thermally-induced repair and reprocessing cycles, thus extending their life span. Both polymeric materials derived from renewable sources and non-toxic (styrene-free/formaldehyde-free) crosslinking systems will be considered as preferential options. In the temperature-enhanced solvolytic demanufacturing process for fibre liberation and resin recovery, industrially relevant (bioderived, when applicable) solvents will be selected, avoiding hazardous substances with critical toxicological profile.

1.2 Methodology

1.2.1 Overall methodology

The proposed methodology in RECREATE involves the cooperation between academic institution, research and development centers, SME recycling companies, technology-oriented SMEs and large industries as well as associations, following a multi-level approach and targeting several technological and non-technological actions.

The Consortium has been carefully built covering all the needed expertise. **Two technological recycling SMEs (COBAT COMPOSITI and Carbon Cleanup) in the project have been involved**. Their role in the project will be the mapping at EU level of existing composite waste, and the selection and supply to other partners of EoL parts (including complex multimaterial FRCs) in suitably agreed geometries/sizes to allow subsequent remanufacturing activities. In addition, they will also be involved in technology and recycled materials validation, benchmarking the RECREATE project results with solutions currently present on the market (e.g., quality of CF fibers from solvolysis *vs*. pyrolysis).

Parts supplied by the recycling companies will be successively investigated with complementary non-destructive techniques (NDT) by IRIS and FRAUNHOFER, in particular by LIBS and active/passive thermography, X-ray CT, ultrasound-based and radar-based technologies (Fig. 7). The objective here will be to understand the nature and composition of complex multimaterial composites and to evaluate their structural integrity in view of their possible reuse. IRIS will develop and build a recognition monitoring prototype unit (TRL6) bringing together LIBS, engineering aspects (e.g., conveyor belts, samples positioning), advanced AI and chemometric techniques embedded in a customized software, for a future use as a smart sorting unit in continuous processes.

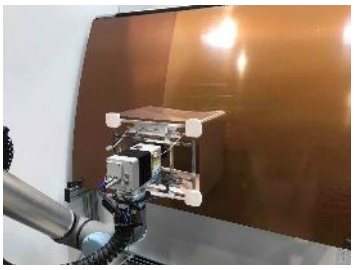

Fig. 7 Active thermography using a collaborating robot (FRAUNHOFER).

An AI-based decision support system (AI-DSS, developed as a digital platform at POLIMI) will serve as digital tool for the selection of the most convenient disassembly/cutting strategy, and of the parts directed to remanufacturing and reuse. In the case of excessive damage and/or degradation of the EoL part, the latter will be instead exploited for fiber liberation and matrix recovery, using innovative technologies aimed at obtaining clean fibers with a minimal physical damage (namely, using catalyst-assisted green solvolysis and electrofragmentation). Both recycling activities will be undertaken at POLIMI. Retention of the mechanical response (modulus, strength) of the liberated recycled fibers will be tested and validated at TAU using a unique, high-sensitivity set-up for single fiber mechanical testing (microbond technique, by Fibrobotics™) available from a TAU spin-off company.

Disassembly, cutting and – in some cases – repair of parts to be remanufactured will be carried out at CNRS through advanced approaches based on high-precision, tunable laser-shock and laser-machining techniques.

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The main strategies and technological approaches used for the realization of the demonstrators will include (Fig.8):

- **<u>Strategy A1: reuse of current generation, EoL GFRC and CFRC parts</u>**, through photonic-based material identification and sorting techniques, high precision laser-based dismantling and machining, NDT for investigation of structural integrity, T-assisted reshaping, design and application of reversible joints;
- **<u>Strategy A2: fiber liberation and reusable matrix recovery</u>**, through catalyst-assisted green solvolysis and electrofragmentation;
- **<u>Strategy A3: design and demonstration of next-gen FRCs</u>** from recycled fibers and smart polymer matrices.

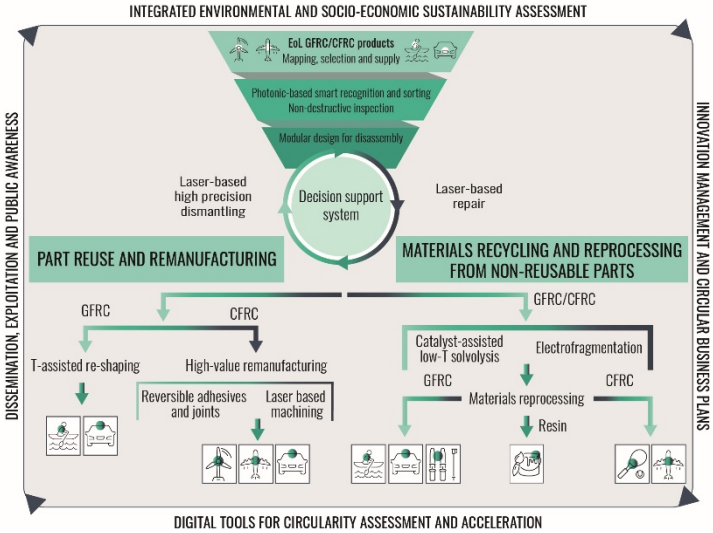

Fig. 8 Reuse/remanufacturing and recycling/reprocessing strategies.

In particular, **different routes for composite reuse** and **innovative recycling technologies** will be validated up to TRL6 through the realization of 9 demonstrators, schematically listed in the following **Table 2**.

Table 2. Demo-case description, key technology involved, leading partner and relevant KPIs.

Demo-case	Strategy / sector	Output product	Key technologies	Leader	KPIs
1	Reuse of GFRC parts for transport	Technical boards	LIBS, NDT Laser shock dismantling T-assisted reshaping	GAIKER	Bending strength 30-40 MPa; Bending modulus > 7000 MPa; Weight < 12 kg/m^2 (6-12 mm thick). 30% reduction of CO_2 emissions
2	Reuse of CFRC parts for mobility	Structural parts for body-in-white in vehicles	LIBS, NDT Laser shock dismantling Reversible joints	EDAG	> 80% crash performance compared to classic design with parts made of virgin material.
3	Reuse of CFRC parts for energy	New concept wind turbines	LIBS, NDT Laser Shock dismantling Reversible joints	REST UG	Full mechanical properties retention; ≥ 50% substitution of virgin fibers; 30% reduction of CO_2 emissions; 20% cheaper/faster repair process (vs. reference rotor blades).
4	Reuse of GFRC and CFRC parts for aviation	Seat back-rest and cabin panels	LIBS, NDT Laser Shock dismantling Reversible joints	GEVEN SPA	8%÷12% weight-saving *vs*. market-available standard; 5%÷10% RC cost-saving *vs*. reference process; 12%÷25% fracture toughness increase *vs*. reference CFRCs. 30% reduction of CO_2 emissions
5	Recycled GF/CF for aviation	Cabin sidewall window panels	Catalyst-assisted solvolysis Electrofragmentation	GEVEN SPA	8%÷12% weight-saving *vs*. market-available standard; 5%÷10% RC cost-saving *vs*. reference process.
6	Smart recognition/sorting systems	Operating sorting/recognition prototype	LIBS	IRIS	CFRC/GFRC composite recognition in waste flows. Flow rate recognition of 25 kg/h.
7	Recycled CF for sport	Padel/pickleball rackets	Catalyst-assisted solvolysis Electrofragmentation Reversible polymer matrices	HEAD Sport GmbH	Fatigue test passed (cycle load of 10.000 ball impacts).
8	Recycled GF for sport	Ski parts	Catalyst-assisted solvolysis Electrofragmentation	HEAD Sport GmbH	Processability on prepreg line: resin uptake > 50 g/m2.
9	Industrial coatings	Anticorrosion primer	Catalyst-assisted solvolysis	BENASEDO	Pencil hardness > B; cross cut adhesion 100%; corrosion resistance > 1000 h salt fog test.

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The aim of the demonstrators (Fig.9) will be the **synergistic integration of all key technologies developed in the project**, which will be **demonstrated in relevant environment at TRL6**. In parallel, a widespread **involvement of companies** in this part of the project will ensure direct relation and immediate relevance with market opportunities.

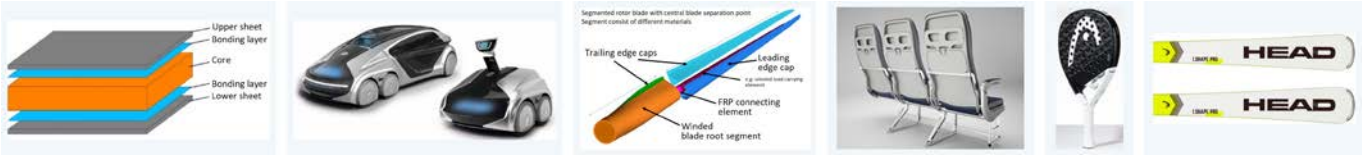

Fig. 9 Representative images of demo-cases developed in RECREATE in WP5 (from left to right: transport, mobility, wind, aviation, sport).

A relevant part of the scope of RECREATE will also be the implementation of **non-technical objectives** aimed at:

- the quantitative evaluation of **socio-economic and environmental performance of proposed solutions**;
- the co-design, realization and validation of **innovative educational and learning instruments**.

More specifically, the methodology proposed and followed in this part of the project is reported in detail as follows.

Strategy B1: Development of new digital tools for circular economy and sustainability assessment and acceleration. The methodology followed will consist in:

- ***development of dedicated digital tools*** such as the Internet of Things (IoT), big data and data analytics to enable the assessment and acceleration of the circular economy principles in six different ways: overcoming logistical roadblocks around raw material collection for reuse and recycling; sorting high-quality feedstock and tracking the quality of the EoL and recycled materials; separating and recycling mixed/complex-material products; enhancing yield and stability of reuse and recycling processes; designing composite parts for reuse and recycling; using emerging digital business models to enable reuse and recycling;
- ***an integrated environmental and socio-economic sustainability assessment*** (ESESA) to address the environmental, economic and social benefits from the reuse, recycling and remanufacturing processes proposed in RECREATE. The ESESA will be based on the following major steps: 1. perform a detailed literature overview on materials and processes to gather data regarding the cost, the environmental footprint and the health-safety effects of the materials used; 2. define the goal and scope of the ESESA for the specific applications and selection of the key technologies to be evaluated from WP2 and WP3; specific requirements to ensure that the processes meet the ESESA will be defined and directed to WP1; 3. define and quantify the functional units to be used in the comparison of the key technologies (in collaboration with the end-users); 4. analyze the overall system which includes the entire life-cycle, manufacturing and the EoL of the demonstrators; 5. perform an inventory analysis at which the material-, energy- and monetary flows of all processes within the system boundaries will be recorded; 6. assess the environmental, social and economic impacts using the SLCA and the LCC methods; 7. perform a final evaluation to compare novel and conventional processes, based on the analyzed sustainability indicators.
- ***the definition of new dedicated circular business models***, associated to two main strategies, namely composite part reuse and composite recycling and remanufacturing. Technology concepts will be verified in a controlled environment designed in tailored innovation workshops organised to engage all participating actors for each strategy, building hypothetical scenarios and designing testing pathways. A circularity assessment tool will be developed. The assessment tool will be based on the review of the innovative technology concepts and on a comprehensive questionnaire aimed at evaluating the degree of circularity of a target business model at each stage of a pre-defined value chain with the respective relevant industrial environments.

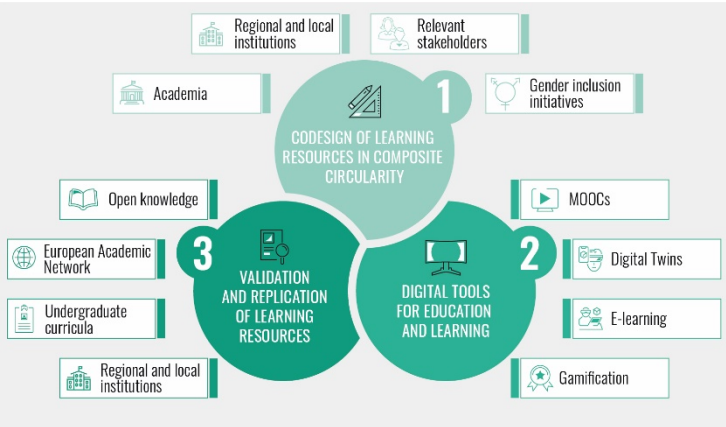

Fig. 10 Co-design strategy of learning resources in RECREATE.

Strategy B2: Co-design, realization and validation of innovative educational and learning instruments through digital technologies. An array of learning resources related to the broad interdisciplinary field of composite circularity will be designed, developed and validated. The learning resources will aim at transferring to current and future generations of employees as well as undergraduate students the core skills needed for the uptake of the resilience solutions developed in the project, with a particular attention to technological processes, as well as social skills, management competences and soft skills, enabling a highly cross-sectorial educational approach. The methodology (Fig.10) will involve:

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- ***the co-design of learning resources*** together with subject matter experts (academic and RTO partners in RECREATE) as well as with representatives of regional institutional bodies (e.g., through the involvement of the Directorate for Research, Innovation and University of Lombardy Region), with partner universities external to the Consortium, with other local educational organizations. This will lead to the production of the RECREATE *"Teaching Methodologies and Strategies"* guidelines. Starting from an in-depth analysis of the objectives of the learning and the specific needs of the different categories of learners identified (current and future professionals, undergraduates), this document will define the active and problem-driven approach to learning, considering also the requirements of modularity and flexibility of the components of the learning resources;
- ***the definition of the digital tools*** to be used to implement such learning resources. The learning resources will be delivered through a gamification platform that will embed them in a specific storytelling, so that learners are stimulated to complete the learning pathways following intrinsic rewarding mechanisms and experiential learning. The use of gamification tools will facilitate the understanding of the key aspects of the subject matters, and support the engagement of learners. The simulation system developed as part of the learning resources will also become a digital twin. The simulator will eventually be able to incorporate open data or output data from real tests, in order to make the simulations even closer to reality;
- ***the validation and replication of such learning resources***. The learning resources will be primarily tested through the MOOC platform managed by POLIMI. The modular characteristics of the learning resources developed will allow to assemble and arrange a number of learning paths with different objectives and target audiences. The MOOC will implement an active approach using a mix of synchronous and asynchronous activities (and tools), such as: video lectures, forum discussions, quiz for formative and summative assessments, single or group project works (asynchronous); theoretical and practical webinars (synchronous). The MOOC will be available on the POLIMI Open Knowledge platform (http://www.pok.polimi.it, currently populated with around 80 courses and 127.000 unique users, and with a specific MOOCs "for teachers"). All the videos produced for the MOOC will also be uploaded on POLIMI Open Knowledge YouTube channel (https://www.youtube.com/channel/UCOKtBGblWKlkE-KIf-S0Fvg), making them widely available to broader community. Finally, the possibility to include such modules in existing or to-be-developed curricula for undergraduate studies will be assessed, with the involvement of some of the European university networks POLIMI and the other academic partners are currently part of (e.g., ENHANCE, IDEA League, Alliance4Tech).

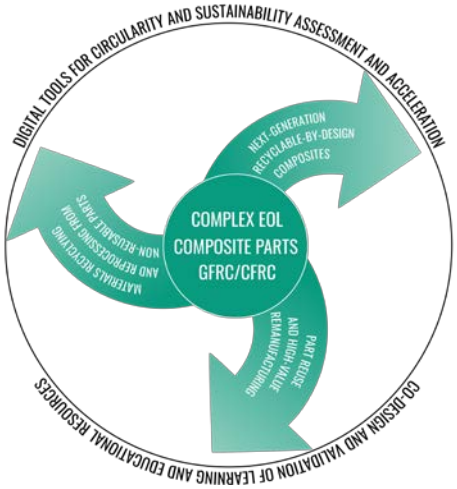

Fig. 11 Overall concept in RECREATE.

The overall concept is schematically represented in Fig.11.

1.2.2 Relation to national and international research and innovation activities

Table 3 lists a selection of relevant ongoing H2020 projects participated by some the Consortium partners and related to the topic of this application. Synergies and cooperation with these initiatives and with other EU projects dealing with materials circularity (including the other projects funded under HORIZON-CL4-2021-RESILIENCE-01-01) will be fostered during RECREATE and implemented within a specific task (T9.4).

Table 3. International funded initiatives related to the main pillars of RECREATE.

List of related projects and initiatives	Summary of topics and exploitable synergies
FiberEUse (2017-2021, ongoing) - H2020, GA 730323. "Large scale demonstration of new circular economy value-chains based on the reuse of end-of-life fiber reinforced composites" *Involved partners*: POLIMI, TAU, FRAUNHOFER, HEAD Sport GmbH, INVENT, EDAG.	The FiberEUse project aims at integrating in a holistic approach different innovation actions enhancing the profitability of composite recycling and reuse in value-added products. *As FiberEUse will act as reference point for circular economy in the composite sector, the partners of RECREATE will interact with FiberEUse to define synergies and exploitation potential in higher TRL phases.*
Alma (2021-ongoing) – H2020, GA 101006675. "Advanced Light MAterials and processes for the eco-design of electric vehicles". *Involved partners*: RESCOLL.	The project will adopt circular economy principles through the application of eco-design strategies to create a novel recyclable and affordable battery electric vehicle platform. *Synergies with RECREATE will be implemented concerning the activities undertaken by RSC and other technology providing Consortium partners, which will find potential alternative exploitation routes on different markets.*
DeCOAT (2019-ongoing) – H2020, GA 814505.	This project will develop a solution to ensure the circular use of textiles and plastic parts with (multilayer) coatings, based on the development of

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"Recycling of coated and painted textile and plastic materials". *Involved partners*: RESCOLL, IRIS.	triggerable smart polymer material systems and appropriate recycling processes. *Synergies with RECREATE will include the identification of new exploitation routes for RESCOLL and other RECREATE partners involved in the development of advanced material solutions for composites circularity.*
MULTICYCLE (2018-ongoing) - H2020, GA 820695. "Advanced and sustainable recycling processes and value chains for plastic-based multi-materials". *Involved partners*: IRIS.	MultiCycle aims to deliver an industrial recycling pilot plant for recovering the components from thermoplastic multilayer packaging and fibre reinforced composites. *IRIS is the coordinator of this project and it is involved in the development of monitoring systems for the fibre reinforced composites, enhancing the competences in fiber recognition area.*
REMADYL (2019-ongoing), H2020, GA 821136. "Removal of Legacy Substances from polyvinylchloride (PVC) via a continuous and sustainable extrusion process". *Involved partners*: IRIS.	REMADYL aims at recycling PVC additivated with hazardous legacy substances (LS) such as phthalate plasticisers (mainly DEHP) and heavy metal-based stabilisers (mainly lead). *In this context, IRIS develops a LIBS system to black PA and PP parts, acting as reference point to identify other black plastic component in RECREATE.*
CarE-Service (2017-ongoing) – H2020, GA 776851. "Circular Economy Business Models for innovative hybrid and electric mobility through advanced reuse and remanufacturing technologies and services". *Involved partners*: FRAUNHOFER, POLIMI.	The CarE-Service project aims at demonstrating Innovative Circular Economy Business Models based on advanced mobility services exploiting hybrid and electric vehicles. Such business models will entail re-use, remanufacturing and recycling of components and materials of hybrid and electric vehicles for applications in the automotive sector as well as in other sectors. *CarE-Service will give important input regarding re-use, remanufacturing and selective recycling of techno-polymeric components*
CASTLE (2016-2022, ongoing), H2020, GA 699777. "CAbin Systems design Toward passenger welLbEing". *Involved partners*: GEVEN SPA, INVENT.	The CASTLE project aims at improving passenger cabin environment by means of an innovative and integrated design approach including both human factors and environmentally friendly materials. *RECREATE partners will interact with CASTLE to define synergies in higher TRL phases.*
FIBREGY (2021-2023, ongoing), H2020, GA 952966. "Development, engineering, production and life cycle management of improved FIBRE-based material solutions for the structure and functional components of large offshore wind enerGY and tidal power platforms". *Involved partners:* AVK.	The overall objective of the FIBREGY project is to enable the extensive use of FRC materials in the structure of the next generation of large Offshore Wind and Tidal Stream (OWTS) platforms. The project will develop, qualify and audit innovative FRC materials for offshore applications. *AVK will serve as direct link with this initiative, evaluating potential synergies especially concerning the foreseen end of life of such novel OWTS structures.*

1.2.3 Interdisciplinarity

The RECREATE Consortium is built upon a vast group of participants from a broad variety of backgrounds, spanning academic and industrial research, higher education, technology development, small, medium and large industrial entrepreneurship, stakeholders' representation, business management and innovation. The **knowledge base is highly interdisciplinary** combining competences in materials science and engineering, industrial manufacturing, product design and engineering, physics and optics, chemical engineering, chemistry, mechanical engineering, business management and implementation, cyberphysics, environmental engineering, education sciences, digital technologies, computer graphics and design. The success of the project relies on the interaction of expertise from different scientific, technological and industrial sectors, with each fundamental step of the project representing a **field for cross-fertilization** as a result of beneficial synergies and interactions among different disciplines. Materials engineering as well as materials manufacturing are *per-se* fields of intersections and intimately require synergetic collaboration between many different fields of knowledge. However, other more **specific interactions and collaborations across diverse yet complementary disciplines** will be required for the achievement of the proposed scientific and technological innovations: i) *dismantling and sorting technologies* – key disciplines: physics, photonics, mechanical engineering, materials and manufacturing engineering; ii) *composite part reuse and remanufacturing* – key disciplines: materials engineering, product design, mechanical engineering, cyber-physics, laser physics; iii) *sustainable and profitable recycling technologies for composite materials* – key disciplines: chemical engineering, production engineering, chemistry, mechanical engineering, process engineering, business engineering; iv) *pilot level demonstration of reuse/recycle approaches* – key disciplines: production engineering, materials engineering, chemical engineering, prototyping, chemistry, manufacturing engineering, business engineering; v) *digital tools for circularity and sustainability assessment* – key disciplines: chemical engineering, environmental engineering, software engineering, business engineering and management; vi) *co-design of learning resources* – key disciplines: education sciences, pedagogy, software engineering, graphic design, computer graphics. Throughout the project, the Consortium will promote the engagement of actors from different scientific and technological disciplines and diverse business sectors implementing **specific actions fostering continuous knowledge exchanges between partners** (see Section 2.2 for detailed measures to maximize impact).

1.2.4 Integration of social sciences and humanities

One of the important objectives of the RECREATE project is the design of innovative learning/teaching resources to enable the implementation of lifelong learning programmes for current employees active in the cross-sectorial area of circularity of composite materials, in addition to targeting the future generation of workers by interacting with local and regional educational organization at undergraduate level. To successfully address these aspects, competences in the fields of **pedagogy, education sciences as well as learning sciences and technologies** are essential. Such skills will be implemented within the RECREATE project through the active participation of the "*Innovation in Teaching and Learning*" task-force (METID) of partner POLIMI. METID@POLIMI specializes in designing and testing tools and methods for learning innovation and faculty development, and will lead the activities related to the definition of innovative learning/teaching methods (T7.1) and to their modular replication and validation (T7.4). These activities will be undertaken in the framework of a WP entirely dedicated to the **co-design and validation of learning resources (WP7)**, led by partner GRIFO whose core digital expertise is in the development and technological implementation of e-learning/gamification solutions at different educational levels.

1.2.5 Considerations on gender dimension

The different activities foreseen in RECREATE will require specific considerations on sex/gender dimension and on how this issue will be taken into account at each level of the project development. In this respect, the Consortium has identified three distinct levels of scrutiny and intervention. LEVEL 1 – PRACTICAL IMPLEMENTATION OF SCIENTIFIC AND TECHNOLOGICAL TASKS PLANNED IN THE PROJECT. The practical implementation of the scientific and technological activities planned in RECREATE is assumed to be equally perceived and participated by male and female subjects and no gender/sex-based differences are expected in the development, demonstration and validation of the proposed technologies. No specific sex/gender analysis related to the execution of such activities is deemed relevant for RECREATE. Should the sex/gender dimension become relevant in this respect as advised by the Consortium partners or externally, appropriate measures will be implemented to ensure full consideration in project activities in line with EC guidelines.[20] LEVEL 2 – DISSEMINATION OF PROJECT RESULTS. The project will take particular care of the sex/gender inclusion topic by designing dissemination material free from discrimination of any kind, by stimulating the participation and involvement of individuals with arbitrary gender orientation in the project activities and outcomes, and by taking specific dissemination actions in synergy with associations and organizations that regularly deal with these issues such as e.g. *Valore D*, Society of Women Engineers, IEEE Women in engineering and Parks. It is widely recognized that in the area of science, technology, engineering and mathematics (STEM) there is still prevalent gender unbalance as well as a large underrepresentation of e.g. women.[21,22] Access to STEM fields both at higher-education/academic level and at professional level is still held back by discrimination, prejudice, social norms, and expectations that make these fields less accessible for women and those with different gender orientations. For these reasons, attention to the use of language in these and other contexts becomes a key element in creating a new vision of society, and particular attention needs to be given to sex- and gender-inclusive language. It is through language that we get an idea of our society and the values we want to uphold. Teachers and trainers need to have a constant and strong commitment in this respect, not only using conscious and consistent language, but also raising participants' awareness and attention to this aspect. Indeed, the model proposed by the teacher constitutes a key starting point for reflection on this matter. Building upon these considerations, the RECREATE Consortium will take full responsibility in disseminating project results using an inclusive language, free from sex/gender stereotypes. The partners will implement dissemination activities using gender neutral language (e.g., avoiding the use of masculine forms/pronouns/nouns in a universal meaning). Particular attention will also be given to the visual language in the creation of dissemination materials such as social posts, newsletters, graphic images or videos, where people/gender diversity and sex parity will be constantly sought and ultimately ensured. LEVEL 3 – CO-DESIGN AND VALIDATION OF LEARNING RESOURCES. The issue of how to develop an inclusive digital language in the broad field of STEM education is crucial for enabling a measurable qualitative and quantitative impact of such disciplines on society. Within this context, POLIMI is actively involved in the FOSTWOM project, a three-year (2019-2022) initiative co-funded by the EC's Erasmus+ for Higher Education program in which the inclusive potential of massive open online courses (MOOCs) is capitalized on to enable STEM subjects free of gender-based stereotypes. The main objective of such MOOC development platform is to raise awareness around gender balance indicators and ultimately enable a more inclusive MOOC design, thus reaching a wider learning audience. Based on this experience, RECREATE will take advantage of the direct involvement of POLIMI in FOSTWOM to co-design teaching and learning resources for lifelong training of current and future professionals in the broad field of composite circularity, following the guidelines outlined in FOSTWOM for inclusive content, free of gender stereotypes. Additionally, MOOCs and digital learning approaches will be employed in RECREATE as strategic means for actively promoting a culture of inclusion. In this context, a representative example is the MOOC "Embracing Diversity" created as part of the IN2IT Erasmus+ project[23] in which POLIMI was involved, now available in open-access mode to interested users on the Open Knowledge portal of

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POLIMI.[24] These previous experiences will constitute the informed background for RECREATE to embrace diversity as key means to develop talents, foster creativity and boost innovation in academia, society and business.

1.2.6 Open science

Open Science practices will be implemented at different levels and different stages during the project to ensure cooperative and systematic sharing of the produced knowledge and tools, in agreement with the IPR protection strategy outlined in Section 2.2.3. In the early dissemination of the scientific results, **pre-print scientific publications** in suitable open access online databases and repositories (e.g., ChemRxiv, arXiv) will be considered for immediate knowledge sharing with the community, thus promoting effective scientific and technological advancement. Additionally, **institutional repositories** (e.g., Re.Public@Polimi, Trepo@TAU) will be used for immediate access to research outputs through univocal persistent identifiers (PIDs), at the latest at the time of publication. Similarly, scientific articles based on the outcomes of the RECREATE project will be published in **open access** mode (either "green" or "gold") and made available to the readership under CC-BY, CC-BY-NC/ND or equivalent publishing licenses. When available from the selected publisher, the **open peer-review** model will be pursued. Beneficiaries and authors will need to retain sufficient IPR when complying with such open access requirements. To this end, sensitive information subjected to IPR protection will be managed according to the IPR strategy outlined in Section 2.2.3, and in the case project results are eligible for patenting, actions will be taken at the earliest possible date. A preliminary agreement has been reached among the project partners to investigate the possibility of opting for the **open-source, license-free and maintainable model for the developed technologies** (including software-based) in order to ensure re-use, maintainability and further development by the entire research and industrial community. Project results that will be provided as open-source components will be protected under open source licenses; analogously, project results that will be commercially exploited will be *copyright-protected*. In order to ensure clear and efficient knowledge management and exchange within the Consortium, the Project Coordinator (PC) will establish internal web resources for communication among the partners and for the dissemination and transfer of knowledge and information as required. All Consortium members will co-operate in order to reach the most efficient process of **knowledge management as well as data management** (see Section 1.2.7). The know-how exchanged between the partners may include, in certain cases, background knowledge. The Consortium has agreed on the principles for the management of IP, as preliminarily set out in Section 2.2.3.

Active engagement of relevant knowledge actors including citizens, civil society and appropriate stakeholders will be continuously promoted throughout the project with dedicated actions, such as the participation in dedicated dissemination events and the organization of tailored dissemination activities (see Section 3 - WP8 and Section 2.2 for a detailed description of the preliminary Communication, Dissemination and Exploitation plan). In addition, specific actions will be implemented in the **co-design of learning resources** for professionals as well as for undergraduate students to promote lifelong learning in the field of composite materials circularity, based on the scientific and technological outputs of RECREATE. To this end, WP7 will be entirely dedicated to these activities, which will be led by partner GRIFO with the direct support of METID@POLIMI, the "*Innovation in Teaching and Learning*" task-force @POLIMI. The activities in this context will follow the recommendations provided by UNESCO for open educational resources (OER) as "learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, reuse, repurpose, adaptation and redistribution by others".[25] In particular, MOOCs will be extensively exploited in RECREATE as an effective strategy to enable broad accessibility to the learning material, with great potential for modular replicability, scalability and world-wide openness.

1.2.7 Data management plan

The Consortium partners have considered issues related to the management of data and research outputs as they want to ensure them to be **findable, accessible, interoperable and reusable** (**FAIR**), ultimately enabling data security, accessibility and reproducibility. A Data Management Plan (DMP) will be implemented as part of a concrete research data management strategy which will minimize possible duplication of effort and the risk of data loss and will offer integrity to the data (first draft at M06, updated at M24/36/48). Furthermore, this approach will enable increased efficiency of research and development activities, time and resources savings in the long term, and compliance with best practices in higher TRL phases. **G_MEDIA as Communication, Dissemination and Exploitation Manager will coordinate the DMP activities** and will take responsibility for the collection, management, sharing and curation of the data, as well as for the day-to-day quality assessment. The DMP will describe plans for creating, organizing, documenting, storing and sharing data. It will take into account issues such as data protection and confidentiality, data preservation and data curation, and will provide a framework that will support researchers and their data throughout the course of their research activities and beyond. It will be periodically updated describing what **type of data** will be generated (e.g., experimental, numerical, graphical) and what policies apply to the data (i.e., **data accessibility**). Specific **unique persistent identifiers** (PIDs) including Archival Resource Keys (ARKs), Digital

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Object Identifiers (DOIs) and Handle System Uniform Resource Names (URNs) will be considered to ensure findability of data and research outputs. **Funding and legal policies as well as data management practices** such as backups, storage, access control, archiving will be defined in the DMP. Moreover, the Consortium partners will clearly describe which data will retain value after the end of the project, how their reuse will be enabled, and how the long-term preservation of such data will be ensured after the original research is completed, to ensure effective **data and research output reusability**. Partners will share the generated research data allowing other partners to replicate, validate, or correct their results, thereby improving the scientific record and ensure knowledge advancement. This will increase research integrity and replication as those who make use of their data and cite them in their own research will disseminate the results possibly in other disciplines, sectors, and countries. Partners will not share sensitive parts of their research data whenever the achievement of the main objectives of the project may be jeopardized by making those specific parts of the research data openly accessible. The methodology and standards, whether and how these data will be shared and/or made open, as well as how they will be curated and preserved, will be defined in the DMP by the Consortium during project implementation. For each dataset generated, the DMP will be based on the following essential implementation steps (Fig.12):

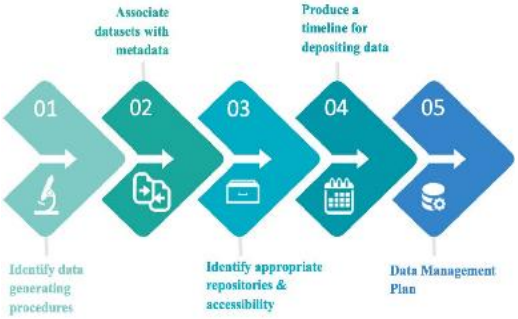

Fig. 12 Data management plan strategy in RECREATE.

- ✓ Data set reference and name: identifier for the produced data.
- ✓ Data set description: description, nature, scale, information about integration and reuse etc.
- ✓ Standards and metadata: reference to the standards that will be used and how the metadata will be created.
- ✓ Data sharing: access procedures and license (e.g. Creative Commons, Open Data Commons), software or other tools needed for access, repository, reasons for confidential data.
- ✓ Archiving and preservation: procedures for the long-term preservation of the data, estimated end volume.

2. Impact

2.1 Project's pathways towards impact

RECREATE overall scale and significance: RECREATE directly affects through its demonstrators 70% of the composite consumption sectors (EU Composite industry market size 16B€ in 2019, growing at 8.8% worldwide in terms of CAGR from 2021-2025),[26] i.e. transportation (14% in value,[27] EDAG), wind energy (14%, REST UG), aerospace (15%, INVENT, GEVEN SPA), consumer goods (24%, HEAD Sport GmbH). Moreover, in terms of geographical distribution of composite use in manufacturing, the RECREATE covers the 90% of the European market distribution. Therefore, there is a massive potential for replication and multiplication of the RECREATE solution at world-wide scale.

RECREATE key challenge and opportunity: The reduction of cost of fibers is the greatest opportunity in the composites market. The reduction in the cost of fibers is expected to decrease the cost of composites, thereby driving a wider penetration of fiber composites in various applications, expecially in the e-mobility sector. At the same time, issues related to recycling pose the greatest challenge for composites manufacturers**.** The complex material compositions and the crosslinked nature of composites, especially thermoset composites, make the recyclability of composites difficult. Most of the composites used are disposed in landfills, or incinerated. As a consequence, by enhancing the greatest market opportunity, i.e. making lower cost re-usable materials available in traceable and certified way, and by limiting the market threats, i.e. demonstrating viable demand-driven solutions for circular business cases, RECREATE will be a *game changer* in the composite industry. Therefore, the economic, social and environmental figures derived in the following paragraphs, focused on a sub-set of sectors due to space limitation, are only conservative estimates of the potential impact of RECREATE at global worldwide scale and at European scale in the next 5-10 years, in line with the European Green Deal time horizon.

Current status and trends of the European wind energy industry. The European wind energy industry (NACE Group 29.1) can be mainly divided into onshore and offshore solutions. In 2019 wind energy supplied 15% of the EU's electricity. In 2019 there were in total 153GW of installed wind energy capacity in the EU-28, corresponding to 117GW onshore and 37GW offshore.[28] By 2030 it could reach 350GW, supplying up to 24% of electricity demand. The European Commission, in the long-term decarbonization strategy to 2050, estimates that wind alone could provide 50% of the EU's electricity demand by 2050. Europe is the main market for wind power generation and the global leader in offshore wind. The global Wind Energy market size is projected to reach 9B€ by 2026, and the European wind power market is expected to register a CAGR of 6.5% until 2026. The outbreak of the COVID-19 pandemic during the initial quarters of 2020 moderately subdued the growth of the wind energy sector.
Small and Medium size wind energy sector: Small-scale wind energy for private use is still in the beginning phases in countries like Germany. These turbines are typically single installations on a farm, domestic property or small business, and they primarily provide electricity for on-site usage. Typical annual energy production is up to 1000kWh, for small turbines with blades up to 18m long, and up to 60000kWh, for medium turbines with blades up to 25m long, with maximum power of 200kW (50kW according to IEC 61400-2). Europe small wind turbine sector accounted for the largest market share of 41.32% in 2016, with a market value of USD 1,440.2 million and is projected to continue to develop growing at a CAGR of 16.5% (global cumulative installed capacity of 4.8GW by 2025). Interest in this technology is gaining traction in light of rising electricity prices and the sociopolitical direction we are heading in to invest more in renewable energies in the future. In order to be able to produce electricity evenly throughout the entire year, the combination of small-scale wind energy, photovoltaics and storage systems is of particular interest. Small wind turbines, for domestic and small-scale commercial use, are currently relatively expensive and struggle to compete with solar solutions. The key to increasing their competitiveness is bringing down their cost and improving performance. The RECREATE circular business case of disassembling blade sections and remanufacture them to be sold at lower cost while keeping the performance of new blades could unlock the diffusion of this solutions especially in peri-urban areas and local eco-systems.

Baseline situation with respect to post-use turbine blades composites. The recyclability of wind turbines has been reported as an industry challenge restricting growth. The standard lifetime of a wind turbine is approximately 20-25 years; the challenge for the materials manufacturers and wind OEMs is to enhance the re-use of materials.

Current status and trends of the European aeronautics industry. The European aeronautics industry (NACE Group 35.3) is one of the key, high-tech sectors of the EU, employing about 794,695 people in 2014, +2.2% with respect to 2013, and generating a turnover of 140.5 billion, with a gross value added of 38 billion euro in 2014. R&D expenditure for aeronautics reached in €20 billion in 2014. The sector in Europe is dominated by the production of large civil aircrafts – aircrafts designed to carry more than 100 passengers. The aircraft traffic measured in RPK (revenue passenger-kilometers) will double in the next 20 years with an annual growth of 4.6%. By regions, Europe's traffic will grow by 3.6% and airlines forecast to acquire more than 7,300 new airplanes valued at over $1 trillion.

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Clearly, this sector and the entire industry has been profoundly affected by the recent COVID-19 pandemic crisis, therefore needing more than other sectors a boost towards innovation to speed-up recovery in the post-COVID era.

Baseline situation with respect to post-use aeronautics composites. 6,000 commercial planes are expected to reach EoL by 2025. Airplanes are typically 50% made of composites. This means around 100,000 Tons of composites from airplanes, with a share of approx. 20% glass fibers and 80% carbon fibers, are expected to be disposed EU-wide.

Current status and trends of the European automotive industry. The automotive industry (NACE Group 34.1) is a strategic sector in the EU, producing 16.2 million cars, vans, trucks and buses every year. A recent report of European Automobile Manufacturers' Association (ACEA) estimated that 12.9 million people - or 5.3% of the EU workforce - are employed direct or indirectly in the automotive sector. The 3.1 million high-skilled jobs in automotive manufacturing represent 10.3% of the EU's manufacturing employment. The transportation sector accounts for the 25% of the GHG emissions according to the IEA (International Energy Agency). In this context, a promising strategy for reducing carbon emissions is the Electro-mobility. It is estimated that a realistic market share for new, electrically chargeable vehicles would be 3% to 10% by 2020 to 2025 and up to 35% by 2040. The key factors contributing to the growth of composite penetration in this market are increased demand for electric and hybrid vehicles, a constant need for fuel efficient cars (reduction in vehicle mass by 10% improves fuel efficiency by 6.5%), no compromise in safety and component mechanical properties, and stringent environment regulations.

Baseline situation with respect to post-use automotive composites The ELV directive requires automotive manufacturers to reach 95% material recovery by 2020. Since recovery routes for composites are currently not in place, the increase in composite share in cars can strongly undermine the achievement of these target limits, especially for future hybrid and electric vehicles. This poses serious concerns on the recovery and remanufacturing of composite parts in the automotive sector. Composite use in the automotive sector is currently 92% glass fiber, 7% natural fibers and 1% carbon fibers. Carbon fibers are mainly used in chassis and fenders, but their cost makes them difficult to be applied at large scale, although they contribute to a reduction of 75% of the part weight.

Pathway to impact

The RECREATE project has defined the following pathway to impact (Fig.13), further described in the next sections.

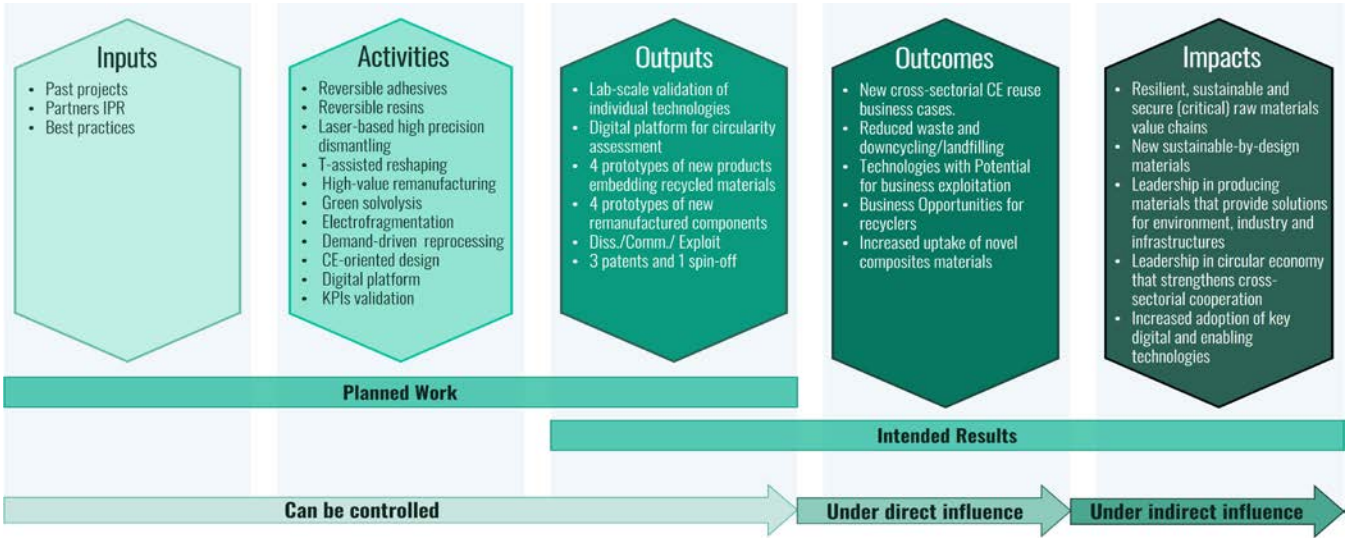

Fig. 13 RECREATE pathway to Impact: from activities to Expected Outcomes [EO] and Wider Impacts [WI].

Basic assumptions and parameters

The calculation of all the quantitative analyses reported in this section is based on the following set of assumptions, data, market figures, extracted by the referred sources.

Parameter	Value used in the quantitative impact forecasts	Source
Avg. price of virgin fibers (varies with type/form).	2300 €/ton for GF and 20130 €/ton for CF. Trend: increasing.	Frost&Sullivan, 2015.
Potential price of recovered and re-usable fibers.	1350 €/ton for GF and 6000 €/ton for CF	AVK-TV Composites Market Report 2019, FiberEUse H2020 outcomes (D2.6).
Cost of processes.	Electrofragmentation: 1200 €/ton; Solvolysis: 9000 €/ton.	RECREATE target KPIs.
Energy and emissions	Electrofragmentation: 1.2 kWh/kg, 770 gCO_2-eq. Solvolysis: 3 kWh/kg, 1196.22 gCO_2-eq.	SELFRAG CFRP Project. Karki et al., 2019, Oliveux et al. 2015.
Expected volume of disposed turbine blades.	In 2025, 70,000 t of composites from rotor blades with a share of approx. 70% glass fibers and 30% carbon fibers are expected to be disposed EU-wide. 10% of the expected thermoset waste in 2025.	WindEurope "Accelerating Wind Turbine Blade Circularity" 2020.

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Expected volume of waste composite parts from aeronautics.	6,000 commercial planes are expected to reach EoL by 2025. Airplanes are 50% made of composites. 100,000 t of composites with a share of approx. 20% GF and 80% CF.	Aircraft Fleet Recycling Association (AFRA). ICAO Environmental report 2016.
Total demand of composite in the automotive industry.	Monetary value of 6.5 B€/year in 2019. A total amount of 2 million tons of GF and 37,000 tons of CF needed worldwide.	AVK market report 2019.
Total demand of composites in the construction industry.	380,000 Tons of GFRP and 6,000 Tons of CFRP.	Composite for Europe, 2017, https://www.composites-for-europe.com/
Average remanufactured product value.	Price for remanufactured part is between 50%-70% of new parts. The regeneration rate in remanufacturing is ~0.8. The average price for a new turbine blade in small wind energy applications (15m long) is ~15k€/part. Remanufactured parts provide higher margin vs. new parts, so estimate margin = 3k€/part is considered.	APRA – Automotive Part Remanufacturers Association. A Detailed Wind Turbine Blade Cost Model, NREL, national laboratory of the U.S. Department of Energy, 2019.
Cost of landfill in Europe.	300 €/ton on average.	Web-sources

2.1.1 RECREATE impact on the expected outcomes

The RECREATE cause-effect diagram towards the expected outcomes is reported in Fig.14. These propagations of effects are further motivated and explained in the following tables, linking the project developments and outputs to the outcomes listed in the call text.

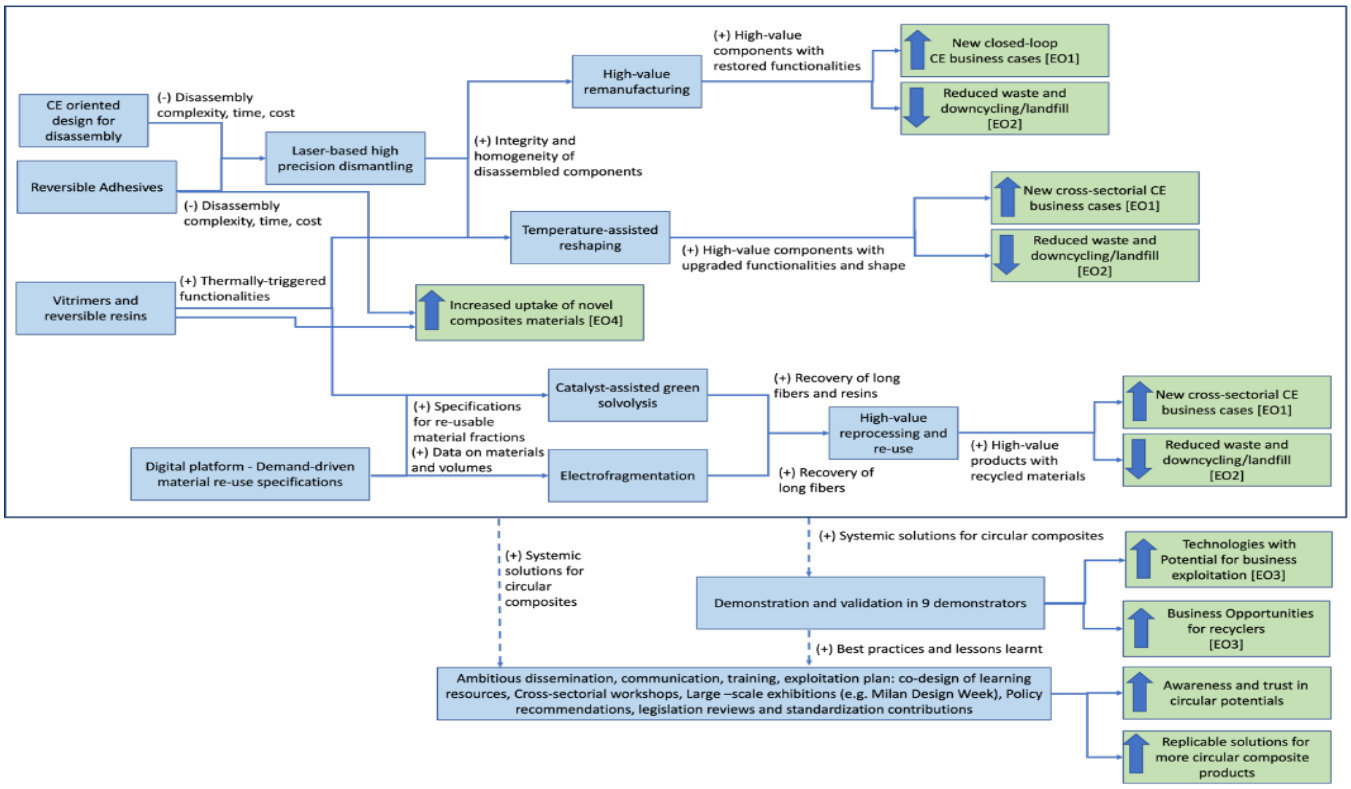

Fig. 14 RECREATE cause-effect diagram - Technological solutions (blue), Expected Outcomes (green).

Expected Outcome 1 [EO1]: Reuse of composite material and recovery of secondary raw materials with higher value than currently available.
Narrative: RECREATE will develop and demonstrate new routes for reuse of composite-made parts including (i) remanufacturing for function restore, (ii) remanufacturing by reshaping and function upgrade and (iii) demand-driven recycling for re-use in high-value products. The first route will be achieved by laser-based high-precision disassembly and reversible joints. The second route will be achieved by thermally-assisted re-shaping after laser-based high-precision disassembly. The third route will be achieved by demand-driven electrofragmentation and catalyst-assisted solvolysis to obtain high-performance fibers and reusable organic fraction. The materials/components obtained by all routes will have considerably higher value than current downgraded uses.
Target Groups: Recycling companies. Manufacturers of high value-added composite parts in different sectors. Composite component manufacturers/remanufacturers.
Scale & Significance: The economic feasibility of the three RECREATE strategies will be validated in 9 demo-cases, validating both closed-loop re-use and cross-sectorial re-use options in real prototypes. The significance will be high since the value of remanufacturer parts will be between the 70% and 80% of the same parts using virgin materials. Moreover, both the electrofragmentation for GFRP parts and the solvolysis for CFRP will feature positive economic performance, with value of the recovered fibers between the 35% (rCF) and the 90% (rGF) of

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the value of new fibers. Transferring the results of RECREATE to the industry, by 2030 a total increase in value added of 0.208 B€ is achievable in the targeted European sectors
Expected Outcome 2 [EO2]: Reduction of waste sent to landfill and positive environmental impact.
Narrative: RECREATE will demonstrate technical and economic feasibility of re-using recycled fractions after selective disassembly by embedding them into high-value products with a cross-sectorial approach. The core idea of the demand-driven vision is that the circular value-chain is transformed into a "pull" system, where the requirements and specifications on the materials and components to be reused are transferred directly from the demand side in terms of characteristics and functionalities of the high added-value products re-using such materials and components. This approach will guarantee that the materials and components obtained through circular economy practices are actually re-usable into new products that are demanded by the market. Moreover, the remanufacturing and recycling solutions of RECREATE will considerably reduce the environmental footprint and energy with respect to current composite production.
Target Groups: Manufacturers of high value-added composite parts in different sectors. Composite component manufacturers/remanufacturers.
Scale & Significance: RECREATE will demonstrate sustainable reuse options for post-use composite parts thus avoiding landfill (90% recovery rates, 80% remanufacturing rate) and reducing environmental footprint of processes (reduction of manufacturing energy: 80%, for GF, 95%, for CF) demonstrated by LCA and energy measurement within demonstrated processes. Transferring the results of RECREATE to the industry, by 2030 a decrease of materials going to landfill of 44,100 kTons/Year (GF) and 7.2 kTons/Year (CF) can be achieved, together with 2.64 TWh/year of energy savings and kTons CO2/year of emissions (see next section).
Expected Outcome 3 [EO3]: Creation of new value streams through new technologies with potential for commercial exploitation; new business opportunities and revenue flows for recycling companies, benefiting particularly SMEs which dominate this sector of the market.
Narrative: RECREATE will bring at higher TRL6-7 innovative promising technologies for remanufacturing (LIBS-based inspection and sorting, laser-assisted disassembly, thermally-assisted reshaping, high-value remanufacturing) and recycling (electrofragmentation, catalyst-assisted solvolysis) that will open new markets both for companies providing these technologies, and for recycling companies, mainly SMEs, that will act as users of these technologies for providing high-value materials to end-reprocessors and manufacturers.
Target Groups: Recycling companies, focus on SMEs. Composite component manufacturers/remanufacturers.
Scale & Significance: Demonstration of circular process-chains enabled by 4 new disassembly and remanufacturing technologies and 2 new demand-driven recycling solutions. Potential sales of new remanufacturing and recycling equipment for about 800 MEuro by 2030. Additional revenue from European recyclers from selling re-usable recovered materials 24MEuro by 2030 (see next section).
Expected Outcome 4 [EO4]: Increased uptake of novel composites materials in industrial applications e.g. enhanced lightweight designs for transport, currently limited due to costs and adherence by industry to environmental legislation and the end of life directive.
Narrative: RECREATE will develop and demonstrate solutions for novel composite materials and components by (i) formulating new smart and green resins, making the composites easier to be recycled and remanufactured, (ii) developing strategies for reversible joining through novel adhesives, and by (iii) designing a new generation of lightweight modular structures for automotive and wind, for easier disassembly, repair / remanufacturing. These solutions will decrease the life-cycle cost of the composite-made components by facilitating multi-use cycles in the same and different (by re-shaping) applications. The reduction of composite part cost will contribute to the wider diffusion of these components in critical industries such as the automotive, contributing to the achievement of targets fixed in the environmental legislation, for emissions during the product use-phase, and the end of life (ELV) directive, in particular in view of the on-going transition to hybrid and electric vehicles.
Target Groups: Composite component manufacturers. Composite material suppliers. Resin suppliers.
Scale & Significance: Demonstration of 2 innovative solutions for reversible joining through activated adhesives, 1 re-design concept for future electric vehicles with modular structures, formulation of new resins for thermally activated processes. Life-time extension of newly design automotive structures of a factor of 5 (target 1.000.000km), reversible resins with disassembly temperature below 150° and disassembly time <30 seconds.

2.1.2 RECREATE impact on long term (Wider impacts-Destination work programme)

The RECREATE cause-effect diagram towards the wider impacts reported in the destination work-programme is reported in Fig.15, showing the connection between individual expected outcomes and the wider impact propagation effect.

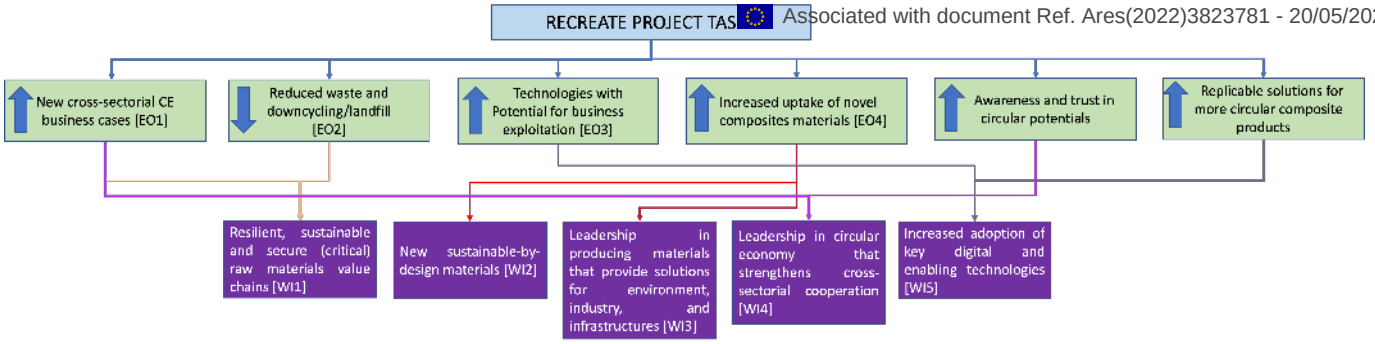

Fig. 15 Cause-effect diagram: contribution pattern of RECREATE to the destination work-programme.

Wider Impact 1 [WI1]: Resilient, sustainable and secure (critical) raw materials value chains for EU industrial ecosystems, in support of the twin green and digital transformations.
Narrative: The improved traceability of material and component volumes and characteristics along the value-chain, facilitated by the RECREATE digital tool, will facilitate the identification of cross-sectorial re-use possibilities in the medium-ling term. RECREATE will increase the resilience and boost socio-economic recovery at local-ecosystem level in the most severely affected sectors, by increasing the availability of secondary raw materials in the local eco-systems by setting up local circular value-chains and by increasing the robustness, adaptability and flexibility of production systems in order to accept both virgin and secondary raw materials. This pathway will boost the green and digital transition of EU companies in different target sectors by providing a concrete scenario to support cost-benefit and financial analyses to properly evaluate the business case.
Target Groups: Local manufactures of composite parts, local recyclers, waste managers, public administrations boosting open innovation, local eco-system facilitators (Clusters).
Scale & Significance: Through the dissemination, exploitation and communication activities of RECREATE at least 30 new re-use business cases will be identified during RECREATE that will be further advanced after the project through local clusters, such as AFIL (Lombardy), Merge (Saxony), Polymeris (France).
Wider Impact 2 [WI2]: New sustainable-by-design materials with enhanced functionalities and applications in a wide range of industrial processes and consumer products.
Narrative: RECREATE will develop and demonstrate solutions for sustainable-by-design materials including by (i) new smart and green resins, (ii) reversible joining through novel adhesives, and (iii) new lightweight modular structures for automotive and wind for multiple use-cycles. Actions towards other industries (e.g. naval, consumer goods, furniture, construction) will be promoted during the project to open new markets to be exploited after the project. The design guidelines and training programs elaborated by RECREATE will make it possible to increase the level of skills of the European manufacturing workforce and to ultimately boost the creation of new knowledge-intensive jobs promoting circular solutions in Europe.
Target Groups: Composite component manufacturers. Composite material suppliers. Resin suppliers. Composite component manufacturers.
Scale & Significance: At least 5 large scale dissemination and demonstration events targeting industries not represented by RECREATE partners (e.g., Milan Design Week). Advanced and innovative learning mechanisms for transferring the technical features of the new materials. Design guidelines for improving the capability of designers to exploit the functionalities of these materials into new consumer products in B2C markets.
Wider Impact 3 [WI3]: Leadership in producing materials that provide solutions for clean, toxic/pollutant free environment, decarbonising industry, and safeguarding civil infrastructures.
Narrative: RECREATE will support the achievement of this wider impact by boosting the capacity of the European industry to reduce emissions, re-use materials, reduce the use of hazardous materials, and avoid waste to be landfilled with direct impact to the automotive, aeronautics and wind energy sectors. *Impact on resource efficient transport system.* The European Commission recently adopted a roadmap, “Towards a competitive and resource efficient transport system”, of 40 concrete initiatives for the next decade to build a competitive transport system. At the same time, the proposals will dramatically reduce Europe's dependence on imported oil and cut carbon emissions in transport by 60% by 2050. These are ambitious targets that may not be concretely reached without composites, fundamental materials for sustainable transport. Composites create a virtuous circle which allows engines to become lighter. RECREATE will support a stable transition to circular economy model for composites in the e-mobility sector, thus boosting their adoption. *Reduced landfill.* According to research published in 2015 in the International Journal of Epidemiology, health is at risk for those who live within five kilometers of a landfill site. Researchers showed a strong correlation between Hydrogen Sulphide (used as a surrogate for all pollutants co-emitted from the landfills) and deaths caused by lung

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cancer, as well as deaths for respiratory diseases, for people living close to landfills. RECREATE will tackle the problem at the source, avoiding waste flows of high-performance materials, thus contributing to improve health and safety of the European citizens. *Reduced use of hazardous substances.* The RECREATE processes will avoid the use of toxic substances which are typical of chemical recycling methods. Moreover, in the remanufacturing stage a particular care will be given to the selection of low environmental impact polymer resins, avoiding, or at least minimizing, the use of hazardous substances like styrene (as reactive diluent) and formaldehyde (as crosslinker).
Target Groups: European citizens at large, manufacturing companies in strategic EU sectors.
Scale & Significance: Massive reduction of composite materials going to landfill (potentially 90% in the long term), market penetration of lightweight composite structure since a solution for their end-of-life will be demonstrated, zero use of hazardous substances and contaminants in the new formulations of resign and composite materials.
Wider Impact 4 [WI4]: Leadership in circular economy that strengthens cross-sectorial cooperation along the value chain and enable SMEs to transform their activities and business models.
Narrative: Through its cross-sectorial re-use approach, RECREATE will simultaneously improve the circularity and sustainability across several sectors in Europe, by providing industry with solutions to keep resources within the eco-system along multiple use-cycles. The "market pull" RECREATE demand-driven circular approach will open the demand for new high-value products re-using materials and components, at affordable prices.
Target Groups: SME in the B2C context, designers, cross-sectorial business facilitators (Clusters).
Scale & Significance: At least 4 new cross-sectorial value-chains demonstrated in RECREATE and at least 15 new cross-sectorial value-chains identified through the open innovation approach of the platform and within the conducted training, dissemination and communication workshops.
Wider Impact 5 [WI5]: Increased adoption of key digital and enabling technologies in industrial value chains and strategic sectors, paying particular attention to SMEs and start-ups.
Narrative: RECREATE will facilitate European high-tech manufacturing SMEs in gaining competitive advantage in new markets by providing them with an innovative set of digital and hardware technologies for demanufacturing, sorting, reprocessing, and re-using materials and components. The RECREATE consortium is very close to SMEs problems, since 8 out of 12 industrial partners are SMEs (67%). In the medium term, the RECREATE solution will translate in a resource efficiency increase worldwide, as the RECREATE systems designed and produced in Europe will be exploited in foreign industries. In this replication and multiplication of effects, RECREATE will also benefit from the existence of a large network of hub for circularity, developed within the "Vanguard Initiative – New Growth Through Smart Specialization", also DG Regio Pilot Action and DG Grow TAF Service, namely "De-and Remanufacturing for Circular Economy", currently involving more than 150 European stakeholders, designed to deliver services to industrial stakeholders by leveraging on multi-regional synergies and complementarities, with the goal of de-risking private investments in innovative and emerging Circular Economy business cases. This initiative will represent a viable solution to upscale RECREATE and promoting a large-scale replication and multiplication of the project results.
Target Groups: European SME, both end-users and technology providers.
Scale & Significance: 8 regional hubs in the inter-regional VI De-and Remanufacturing for Circular Economy network, more than 100 de-risking project expected in the next 5 years, potential turnover of about 500MEuro generated for European SMEs using the hubs for exploring their transformation path towards circular economy.

In the following, the RECREATE consortium has quantified the economic, social and environmental impact of the project for a sub-set of demo-cases cross-linking different sectors. The considered scenarios include the re-use of CFRP from the aeronautic sector back to the aeronautic sector (demo-case 5), the re-use of GFRP from the wind energy sector into the automotive sector (demo-case 6), as well as the remanufacturing of sections of modular wind blades in the small wind energy sector (demo-case 3). The resulting figures are, therefore, conservative measures of the overall RECREATE impact. Data/assumptions are available in *Basic assumptions and parameters* part.

Economic Impacts (overall figures in Table 4)

Economic Impact for the EU wind energy industry. RECREATE will contribute to the wind energy industry by:
• Providing solutions to re-use glass fibers from the wind-energy industry into automotive industry, avoiding landfill.
• Providing solutions to remanufacture and re-use small wind turbine blades.
In 2030, 70,000 Tons of composites from rotor blades with a share of approx. 70% glass fibers and 30% carbon fibers are expected to be disposed EU-wide. By applying the RECREATE solution, the GFRPs of the rotor blades would be recovered and reused into automotive structures. This will result in landfilling cost savings of **13,230,000€/year** by 2030 with a recovery rate of 90%, and, more important, saving 44,100 Tons per year of GFRP going to landfill. The impact of the second innovation is analyzed next. The cumulative small wind energy installations in EU are expected to reach 4.8 GW by 2025, meaning that in 2025 approximately 114,300 small blades will be produced (4

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blades per turbine), with a CAGR of 16.5% and being first small wind installations dated back in 2003. Considering a market share of the solution of 30%, it can be assumed that out of these blades the RECREATE remanufacturing solution will be applicable to 34,290 parts per year, with a remanufacturing regeneration rate of 0.8. This means an increase of value added of ~**82,296,000€/year** by 2025. Summing both, an economic benefit of **95,526,000€/year** is likely achievable in wind energy industry, with total GFRC saving in material to be landfilled of **44,100 Tons/year**.

Economic Impact for the EU aerospace industry. By applying the RECREATE solution, the CFRP composites of airplanes can be recovered and reused in high-added value applications back into the aeronautics sector. By considering a market share of the RECREATE solution of 10% in 2030, with a recovery rate of 90%, an increase of value added for the sector of **80,136,000€/year** can be achieved, also saving **7,200 Tons** of aeronautics CFRP waste being landfilled, with additional, but marginal, cost savings (2M€).

Economic Impact for the EU automotive industry. RECREATE will contribute to the automotive industry by providing solutions to re-use glass fibers from the wind-energy industry. Considering that the EU28 represents about 20% of the total worldwide car production, the demand for glass fibers in the European automotive industry is 450,000 Tons per year. Assuming a market share of the RECREATE solution of 10% by 2030 in the EU28 automotive industry, approximately 45,000 Tons of recycled glass fibers per year may be processed. As 44,100 Tons of GF will be originated yearly from the wind energy industry, by establishing a cross-sectorial value-chain a stable circular economy business equilibrium for the supply and demand of recovered composites can be generated. This would mean a reduction of material cost of about **41,895,000€/year** by 2030.

Table 4 Summary of RECREATE impact on market value in target industries.

Industry	Market share of RECREATE solution in the EU in 2030	Increase in value added in 2030 [B€]	Circular economy output to same/other industries (Ton/year)	Circular economy input from same/other industries (Ton/year)
Automotive	8% (GFRP and CFRP re-use)	0.041	N/a in the considered scenario	44,100 GFRC
Wind Energy	30% (reman)	0.095	44,100 GFRC	N/a in the considered scenario
Aeronautics	10% (CFRP re-use)	0.08	7,200 CFRC (to aero)	7,200 CFRC (from aero)
Total	/	0.203	44,100 GFRC; 7,200 CFRC	44,100 GFRC; 7,200 CFRC
1€ invested in RECREATE by the EU will translate into at least 21,75€ of additional added value for EU industry by 2025.				

Social Impacts (overall figures in Table 5)

Impact on jobs and growth in the targeted European industrial sector:

- ***EU Wind Energy industry:*** RECREATE will increase the added value of the EU wind energy sector by 95.52M€ after the project (2030). Currently, 253,000 people work within the wind energy sector, with an added value generated per worker of 40,700€. This means additional 2347 new workplaces needed by sector after RECREATE.
- ***Aeronautics industry:*** The value added per employee is 78,000€. This means additional 1027 new workplaces that will be required by this industry after RECREATE, since an added value of 80M€ will be brought.
- ***Automotive industry:*** RECREATE will increase the added value of the EU automotive sector by 41,895M€ after the project. According to the actual contribution to the value added per employee in this sector (54,000€), this means 775 new workplaces that will be required by this industry by 2030.

Table 5 Summary of the RECREATE impact in the target sectors in terms of new jobs.

Summary of RECREATE impact on occupation levels in target industries		
Industry	Employees	Increase in employment triggered by RECREATE Projection for 2025
Automotive	12,000,000	775
Wind Energy	253,000	2,347
Aeronautics	487,180	1,027
Total	12,740,180	4149
2410€ invested by the EU will produce 1 new job in these industries at the end of RECREATE		

Impact on raw materials-environment in the targeted European industrial sectors (overall figures in Table 6)

Energy Consumption in manufacturing operations*:* The process for producing composites is strongly energy intensive. In order to produce virgin carbon fibers approx. 165 kWh/kg of energy are needed, while the production of new glass fibers takes an input energy of approx. 50 kWh/kg considering the material production as well as the manufacturing phase for both composites. A typical production process requires, in addition to the energy needed for the fibers, 15 kWh/kg. The liberation of re-usable glass fibers using the RECREATE solvolysis (for CFRP) and electrofragmentation (for GFRP) approaches is estimated to have an energy consumption of 3 kWh/kg and 1.2 kWh/kg respectively. The remanufacturing and repair of carbon reinforced plastics parts is expected to save 90% of the energy with respect to the production of the same new part (based on APRA Europe market figures).

RECREATE re-usable materials: Given these figures, under the result that EU-wide 44,100 Tons/year of post-use glass fibers and 7,200 Tons/year of carbon fibers are re-usable from rotor blades and airplanes in 2030, 2.55 TWh can be annually saved by re-using fiber composites of blades and airplane parts with a cross-sectorial approach.

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RECREATE re-usable components: Assuming 34,290 small turbine blade sections are remanufactured in 2030 a total amount of 0.089 TWh can be annually saved by the new RECREATE remanufacturing process.

***CO_2 Emissions**:* In the production of composites approx. 95% of the energy consumption is linked to direct fossil energy and approx. 5% is linked to electrical energy. For fossil energy approx. 700 g CO_2/kWh, and for the electrical energy mix approx. 500 g CO_2/kWh, are emitted in average. As a consequence, 1.492 kTons of CO_2 can be saved by re-using 44,100 Tons/year of GF from wind turbine blade and 7,200 Tons/year of CF from aeronautics. Moreover, 0.063 kTons of CO_2 can be saved by remanufacturing blade sections in the wind energy industry.

Raw material use in manufacturing operations and reduction of waste going to landfill: In the considered scenario, 51.3 kTons of waste composites per year, that would go to landfill, will be saved by RECREATE (Table 6).

Table 6 Summary of the RECREATE environmental impacts by 2030.

Summary of Environmental KPIs- RECREATE	Savings in materials going to landfill [kTons/year]	Savings in Manufacturing related Energy [TWh/year]	Savings in CO_2 emissions [kTons CO_2/year]
Total	44.1 GF, 7.2 CF	2.64	1555

Scientific Impacts

The technical work of the project involves a wide range of multi-disciplinary tasks, which require partners with complementary skills and backgrounds. RECREATE integrates knowledge from different domains of research:

Disassembly processes	NDT inspection	Artificial Intelligence
Demanufacturing processes	Monitoring and sorting	Cyber-Physical Systems
Remanufacturing/Reprocessing	Design for disassembly	Feed-forward Control
Polymer technologies	Product Life-cycle Modeling	Decision Support System
Materials science	Social Innovation and Inclusion	System and network optimization

This interdisciplinary approach of RECREATE requires a unique combination of skills that can only be provided by the best researchers and professionals of various European countries. The project will therefore strengthen an R&I&D ecosystem that can foster world-class European manufacturers supported by a knowledge triangle of academia, research and industry, working in synergic way towards a cross-KET solution for a Circular Economy in EU.

Potential Barriers and Obstacles

The existing potential barriers and obstacles that could undermine the achievement of the aforementioned impacts are described in the following Table 7.

Table 7 Obstacles and barriers in RECREATE

MAIN BARRIERS & OBSTACLES	ACTIVITIES TO OVERCOME OBSTACLES AND ACHIEVE THE EXPECTED IMPACT
Economic feasibility of the RECREATE demand-driven circular value-chain solution and business model	The value-chain orientation of the project will allow a smooth and fair distribution of investments and risk among different stakeholders. A detailed validation of the economics of each business scenario will be carried out, under different market scenarios.
Environmental benefits of the proposed circular routes	The innovative cross-sectorial circular routes in the RECREATE demo-cases will be analyzed during demonstration by LCA/LCC methods in order to validate the environmental impacts assessed in the proposal.
Reluctance to new technologies introduction.	Gradual introduction of new technologies in the end users process. Training courses to improve the skills of the operators/engineers in the new technologies (WP7). Specific change management practices (WP8).
Replicability of the RECREATE solution.	The flexibility and adaptability of the RECREATE solution, verified in a comprehensive set of demo-cases covering all sectors of relevance for fiber-reinforced plastics, will assure its cross-sectorial application.
Industrial secrecy can influence results and uptake beyond the project.	The partnership agreement must outline the necessarily fine balance between conflicting interests. Identified results have already a designed exploiter since the proposal phase, so partners agree on exploitation roles.
The presence of competing companies can affect developments	This issue has been addressed by clearly separating roles of potential competitors (see consortium complementarity section below). Potential issues will be solved in the dedicated IPR management task (T8.4).
Industrial reluctance to share data can influence adoption of digital platform.	The data management policies designed by the RECREATE FIRE platform will increase the company confidence level and trust while loading on the platform potentially confidential data.
Multi-disciplinarity challenges of the RECREATE approach.	A well-structured and transparent management structure will ensure correct communication among partners and synchronise their activities; several partners have already multidisciplinary competences.
Legislation boundaries on implementation.	RECREATE has already investigated potential legislation barrier that could undermine the project success. The demo-case leaders did not, however, identified serious barriers to the development of the demonstrators.

2.2 Measures to maximise impact - Dissemination, exploitation and communication

The RECREATE strategy for *Communication, Dissemination & Exploitation (CDE)* aims at supporting the achievement of the Expected Outcomes, within the project lifetime, and facilitating the achievement of the medium-long term Wider Impacts, beyond the project. The communication activities will generate awareness of the project among different stakeholders and interest in obtaining the disseminated results. The management of research data and IPR will guide the dissemination activities to serve the academic community while protecting the consortium's interest in commercial exploitation. Dissemination and exploitation of the project results will be based on large-scale demonstration action of cross-sectorial circular value-chains for the recovery and re-use of FRC into high added value products. Among the main existing barriers, D&E actions will address (i) the lack of circular business models for boosting profitability; (ii) the poor customer acceptance for remanufactured products. Main goals of the CDE

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workpackages (WP7-8) will be (i) facilitating the industrial uptake within local eco-systems; (ii) de-risking future industrial uptake through private investments; (iii) increasing awareness for a more conscious consumer behaviour. Large-scale replication and multiplication will take advantage of the Vanguard Initiative "De-and Remanufacturing for Circular Economy" pilot network.

2.2.1 Exploitation of the project results

The CDE plan and the related process will be led by **G_MEDIA**, the RECREATE CDE Manager (Section 3). The company has consolidated expertise in engaging, training and support capacity building with relevant stakeholders brought to the dedicated unit Circular Business Academy (www.circularbusiness.academy) and to the development of the Roadmap Towards the Circular Economy in Slovenia.

In the implementation of the exploitation part of the plan (WP8), the RECREATE exploitation team will coordinate the partners and define the organisational aspects needed to address the market. The process will be based on 4 workshops, namely "WS1: Industrial thematic workshops with potential stakeholders", "WS2: KER exploitation workshop and dedicated exploitation session", "WS3: Exploitation implementation workshops", "WS4: How to pitch" workshop". These workshops will provide each demo-case with the opportunity to work on (i) the identification/grouping of key exploitable results; (ii) the first definition of the related use mode; (iii) the identification and mapping of risks related to the exploitation; (iv) follow-up actions. These WS will follow the path tracked by the Lean Canvas approach that will be used for the Business and Use Model Development of the KERs. A crucial step for exploitation is the identification of the risk factors hampering future use, the evaluation and prioritization of the risk factors and the identification of actions to minimize their impact. Risk factors will be screened following these categories: Partnership, Technological, Market, IPR, Environmental and Financial risks.

RECREATE aims at fostering competitiveness and growth and increasing benefits to the partners involved and to EU economy and citizens. Depending on the project result type, the consortium has developed an innovative method for identifying KERs in a comprehensive and systemic way, targeting different exploitation routes customized to the specific demonstration strategies. Moreover, the RECREATE consortium has analyzed all the exploitable solutions generated at value-chain level and identified four stakeholder groups for exploitation:

- Technology and B2B-service providers – **TP -** and Design & Engineering groups – **DE -**, which are interested in selling the solutions (or the use of the solutions) and new designs developed in RECREATE to largest possible target end-user groups, contributing to a sustainable transition to Circular Economy.
- Recyclers and Remanufacturers – **RR** - (Lead-users), recycling companies and remanufacturing/repair companies interested in using the new demand-driven recovery and remanufacturing solutions developed by RECREATE for (i) selling recovered components/materials to end-users (ii) selling services to end-users.
- Product Manufacturers and B2C-service providers – **MFG** - (End-users): The manufacturers, which are interested in making a concrete transition towards circular economy business models which can enhance the ability to decrease the costs of composite materials, to propose new customized "green" products to the market, and to contribute to the environmental sustainability of their production processes.

According to the project goals, the main exploitable solutions, and the respective exploitation plan will be as reported in Table 8.

Table 8 Preliminary identification of KERs and exploitation routes.

Exploitable products	Main Exploiter(s)	Route to exploitation		Time to market	Target Industry
		During RECREATE	After RECREATE		
New products offered at value-chain level					
Added-value, customized products re-using fibers and resins.	TP: POLIMI, GAIKER, TAU, IRIS. RR: Carbon Cleanup, COBAT COMPOSITI. MFG: GEVEN SPA, HEAD Sport GmbH, BENASEDO.	RECREATE will test the effectiveness of the developed solutions and will demonstrate the benefits at pilot scale in 5 demo-cases for high-value customized products.	The RECREATE demo-cases will be translated into new products. Extensions to other markets will be considered according to the output of the exploitation strategy.	4 years after the project ends. Form: internal product development, RTOs spin-offs.	• Furniture • Automotive • Sport equipment • Design • Appliances. • Consumer goods
FRP components modular design, reman., repair, reforming, and re-use	TP: Fraunhofer, ICAM, CNRS, IRIS, INV, POLIMI DE: REST UG, EDAG. RR: COBAT COMPOSITI MFG: EDAG, REST UG, GEVEN SPA	RECREATE will bring the modular design, disassembly, remanufacturing and reforming processes at TRL 4-6 and will test the properties of the remanufactured components.	MFGs have strategic interest in opening new markets for this attractive circular economy solution, after validation by customers and business model refining.	4 years after the project ends. Form: patenting.	• Wind energy • Marine • Aeronautics • Automotive • Furniture
New sustainable-by-design composite materials with enhanced functionalities	TP: POLIMI, INVENT, EDAG. DE: EDAG. RR: COBAT COMPOSITI. MFG: EDAG, REST UG	RECREATE will develop and demonstrate solutions for sustainable-by-design materials including by (i) new green reversible resins, (ii) reversible joining, and (iii) modular structures.	These solutions will be transferred to the industry at large scale through dedicated dissemination workshops, in order to investigate potential market uptakes in different sectors.	4 years after the project ends. Form: patenting.	• Wind energy • Marine • Aeronautics • Automotive • Furniture

New software technical enablers. MULO (Make, Use, Licensing, Outsourcing)					
Circular Economy digital platform for supporting the adoption of products re-using materials and components	M, L: G_MEDIA, UPAT, GRIFO U: All project partners .	RECREATE will design and develop core data structure and services to support value chain integration and collaborative development of new products using composite.	The RECREATE partners will have strategic interest in promoting the platform and push for the replication, by large system integrator and services providers, in different markets.	2 years after the project ends. Form: packaging of services of the platform.	• All sectors of RECREATE • Customers, • Investors, • Waste mgmt. consortia. • Municipalities. • Regions
Digitalisation of knowledge about new materials, products and innovative processes for composite re-use for training tools and gamification.	M, L: POLIMI, GRIFO U: all stakeholders, including end-customers, service providers, policy makers, regulators, standardization and certification bodies.	RECREATE will develop a large pool of digitalized knowledge, including data, pictures, KPI figures, validation data, description of materials, designs, testing, certifications etc.. for all the demonstrated products and best practices on how conduct a successful replication.	POLIMI and other scientific partners will continuously update the digital platform database and the set of collected examples and will provide the access to the library to the EU stakeholders. GRIFO will design services to make the digital learning tools available to external users.	1 year after the project ends. Form: Licensing.	• Composite production. • Composite re-use industry. • Public authorities, certification and standardisation bodies

2.2.1.1 Preliminary Business Plans for RECREATE Exploitable Products and Enabling Technologies

***Individual Partner Business Plan* – HEAD Sport GmbH:** The main industrial interest is to validate the re-use of CFs in padel and pickle-ball rackets (Table 9). The flat surfaces require isotropic properties and the use of recycled material from early beginning of these growing markets will set the standards. Yet the technical requirements of these impact sports remain high and the material needs to comply with them. The result must be highly competitive against current solutions under development. It is assumed the lower costs of the material allow additional production steps necessary for working with recycled fibers. For recycled GFs, the interest is to replace a fleece with rGFs, which is in nearly every ski produced. The fleece must provide interlaminar strength. To achieve this, there is a potential rGF use of 20-25% of total GF volume used. The introduction of recycled materials should not exceed current target cost.

Table 9 Potential economic benefits for HEAD Sport GmbH demo-cases.

Benefits	RECREATE	+1 years	+2 years	+3 years	+4 years	+5 years
Sales skis [n° units]		50.000	100.000	100.000	200.000	400.000
Sales rackets [n° units]		20.000	20.000	60.000	90.000	100.000
Gross Sales Revenue [M€/year]		7.900.000	1.5400.000	16.200.000	31.800.000	62.000.000
Accumulated Turnover in 5 years						
Expenses [M€/year]	0	7.310.000	13.860.000	14.580.000	28.820.000	55.800.000
EU grant [M€/year]	319.000	0	0	0	0	0
Own investment (M€/year)	120.000	200.000	0	0	200.000	0
NET Benefits per year [M€]	439.000	390.000	1.540.000	1.620.000	2.980.000	6.200.000
Accumulated earnings in 5 years	**12.291.000**					

2.2.2.2 Industrial Impacts: Innovation for Market Success

RECREATE will produce multiple impacts in the European industry. The participation of industrial partners will ensure that the project will directly impact on the following wide range of industries: Aeronautics, Automotive, Wind Energy, Sport Equipment, Marine, the recycling / remanufacturing and the composite manufacturing industry as a whole. The following table summarizes the relevant exploitation strategies of a sub-set of project partners (Table 10).

Table 10 Main strategic interest of partners towards exploitation

End users	
HEAD Sport GmbH	Paddle and pickle ball are young trend sports and grow at a fast level and in new countries. A successful democase will be further developed into serial products, which will probably need two years for market introduction. It is excepted that at least half of the range can be produced by the new technology (appr 5 years time frame), bringing the total revenue >3 Mio Euro.
REST UG	The design and production of rotor blades with recycled GRP material will be set up for a novel 100 kW twin rotor wind turbine. It is expected that the benefits of this new blades will improve the market position of REST UG 100 kW turbine. The technology will also be used to be licensed to other blade manufacturer.
GEVEN SPA	The recycling of seats that contain composites components at present is done by GEVEN SPA customers. GEVEN SPA can exploit project results by adjusting its business model and adopting the developed technologies to recover, recycle, reuse or resale the seats composite components. The investigation on business model changes and how this affects the present GEVEN SPA economic activities (also the financial viability) will be assessed. Technological scale-up at plant level will be implemented in one year after the project with the ambition to have at least 50% more sustainable products within 2030 that will enhance the enterprise sustainability brand and will allow to increase the turnover (estimated around 30-40%), create new employment (+30%) and attract the best talents.
EDAG	EDAG acts as a multiplier and enabler to the automotive OEM and the large system suppliers. EDAG will investigate and try to implement the technology of reusing End-of-Life (EoL) composite waste as a source for parts in different series development projects of automotive body structures (e.g. body in white, interieur, exterieur). Our customers increasingly

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	expect sustainable solutions in car development, and the expected results of this project will be a good lever to offer them what they are seeking.
BENASEDO	BENASEDO will exploit internally the results of the demo-case focused on the re-use of resins for coating applications. A new product line may be launched at the end of the project, if the results are demonstrated to match target KPIs
Technology Providers	
INVENT	Adhesive bonding is important for composite assemblies, but SoA technologies don't allow any disassemble without damage of the components. With a damage free debonding technology INVENT would be able to offer customers novel design opportunities for sustainable products in terms of e. g. recycling, repair, and maintenance. It is expected to develop new business opportunities like follow-up projects and orders from the industry bringing a revenue of >1-2 Mio EUR in the first 3-6 years after the project.
IRIS	Key Exploitable result: Smart recognition and monitoring system for fiber reinforced composites. Type of IPR protection: Patent. Type of exploitation route: commercialization, direct sales. Target users or clients: Waste treatment plants, recyclers.
RESCOLL	RESCOLL will mainly benefit from the licensing of the laser shock technology to companies in the composite industry and other related sectors that could benefit from the continuous dismantling of multimaterial parts. RESCOLL will also benefit from sells of debondable adhesives to industries such as energy and transports that are in demand of such technology.
GRIFO	GRIFO expects to exploit the project results to enlarge its portfolio of e-learning contents and digital games, by adding a new field of application, the promotion of best practices connected to reuse of materials and the circular economy. The direct target for this type tools will be academic entities and public bodies, with a view also to B2C commercial exploitation.
Post-use products (waste) managers and recyclers	
COBAT COMPOSITI	COBAT COMPOSITI will exploit project results i) developing an optimal reverse logistics network configuration for composite waste, and ii) contributing to the definition of actions to mitigate/remove barriers related to the legislative framework. The project outcomes will provide information to support the development of an industrial initiative to be realized in synergy with the Next Generation EU program in order to treat volumes of waste wind blades that will be produced in the coming years in Italy.
Carbon Cleanup	Carbon Cleanup will further develop the recycling strategies for carbon composites together with HEAD Sport GmbH and other potential customers. It is expected that Carbon Cleanup can grow its technology competences into further business models.
Academic Partners and RTD	
TAU	TAU will use the RECREATE results for specific use cases in the future to benefit further research and development actions but also the direct company projects. Further, as an academic partner TAU will focus on using international open-access publication channels to distribute basic knowledge related to recycled fibres enhancing the exploitation of the RECREATE results globally.
POLIMI	POLIMI will exploit the demand-driven approach for composite fibers and resins re-use within the network of industrial stakeholders of the infrastructure for Circular Economy of the Lombardy Region (AFIL Cluster). Furthermore, the research and innovation results achieved in the development of the new recycling and reprocessing technologies will be exploited through patents and technology transfer to local companies within POLIMI's network. POLIMI will also develop and exploit new e-learning methodologies for education, to be further shared with other EU academic partners.
Fraunhofer	Fraunhofer IWU and WKI as well as Fraunhofer headquarters in Munich actively support IP transfer, licensing and the creation of spin-offs and fostering entrepreneurship skills among Fraunhofer employees. Fraunhofer research outputs of this project will be a methodology for design of reversible joints and a methodology for modular design of next-gen composite structures. Furthermore, Fraunhofer will exploit a new process for automatic evaluation, repair and remanufacturing of rotor blades.
ICAM	Disseminate information in the industrial network of Icam and help developing the industrial sector.
UPAT	UPAT will develop two digital tools in RECREATE: a digital tool integrating IoT, big data and data analytics for the assessment and the acceleration of circularity of composites parts and a digital tool for the implementation of the ESESA. The knowledge that will gained from RECREATE will be used in Training Schools targeting scientists and PhD students, in the development of future collaborations and projects and in the updating of the education material in the area of composites sustainability.
GAIKER	GAIKER will transfers or licence this know-how to companies interested in remanufacturing and reuse composite waste into new composite parts in different sectors (like composite panels for transport, energy, or construction). Additionally, the temperature-assisted reshaping know-how will open more opportunities for research with different chemistries composites than polyester increasing the GAIKER research competitiveness.
CNRS	Project Outcomes will provide specifications going towards industry through partner RSC. Produced patents will be licensed. CNRS will organize Training seminar in hybrid laser processes.
Associations	
APRA Europe	APRA Europe represents the remanufacturing industry and is constantly striving to communicate new ideas, technologies, products and solutions as well as precise business cases. The remanufacturing, and, much earlier in the life cycle, the design for remanufacturing, of composite-based products is still not part of current daily operations in the industry but it offers huge potential to be exploited.
AVK	AVK is one of the largest composite networks in Europe. One of the main goals of the AVK is to broaden knowledge about the materials and to promote their use in industry. The knowledge gained in the project serves to support the essential goals of the AVK work. The knowledge gained will not be used in an economic way, but rather used to promote the use of the materials and to make the latest knowledge available to the network.

2.2.2.3 Impact on standardisation (at national and international level)

Standards are important levers when dealing with exploitation in the circular economy area, which nowadays lack of systematic approaches, as well as in relatively new areas such as data management, communications, security and cloud computing. During RECREATE activities, partners will adhere to existing standards (Table 11) as much as possible and will identify the emergence of needs for new standard within WP8.

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Table 11 RECREATE: Identified reference standards

Reference standards, protocols, systems	
Composite standards	ASTM's composite standards are instrumental in the evaluation and determination of the physical, shear, tensile, flexural, and compressive properties of various forms of composite materials used in structural applications. D7205 / D7205M - 06(2011) and D4762 - 11a (Standard Guide for Testing Polymer Matrix Composite Materials). ISO 1268 (Composite Manufacturing methods). http://www.iso.org/iso/iso_catalogue/catalogue_tc/catalogue_tc_browse.htm?commid=49462.
Remanufacturing	ISO/DIS 10987-2 (Earth-moving machinery -- Sustainability -- Part 2: Remanufacturing).
Connection to data sources	OPC-Unified Architecture (Open Platform Communications, industrial Machine to Machine communication protocol for interoperability); Classic OPC; XMPP (Extensible Messaging and Presence Protocol for message-oriented middleware based on XML); MTConnect (manufacturing industry standard to facilitate the organized retrieval of process information from numerically controlled machine tools); B2MML(Business To Manufacturing Markup Language: XML implementation of ISA-95 standard); QIF (unified XML framework for computer-aided quality measurement systems)
Cloud/digital systems	IEEE Standards Association (IEEE-SA) draft P2301 (cloud profile), P2302 (intercloud). EU ETSI Cloud Standards Coordination (CSC) initiative outputs. ISO/IEC 19086-1 (Cloud computing - Service level agreement (SLA) framework), ISO/IEC 20933:2016 (Distributed Application Platforms and Services (DAPS).

2.2.2 Dissemination and Communication of RECREATE results

The Communication and Dissemination part of the CDE plan will tackle both B2B and B2C relations along the cross-sectorial value-chains to reinforce the engagement of participating actors and establish the "pull" effect created by the demand-driven orientation of the project. Both plans will aim at: (I) identifying target stakeholders, (II) articulating modalities to reach and engage them effectively for the project scope, while reducing overlapping with other European projects and initiatives, (III) increasing acceptance levels of products embedding re-usable materials and components by industries, consumers and citizens at large, thus unlocking the future demand for circular product use, and (IV) increasing acceptance level of modern non-ownership business models through cross-sectorial value chains, favouring take-back of post-use products by the consumers and enabling a steadily evolving collection process. Dissemination activities include (i) target groups mapping and analysis, objectives, means, activities; (ii) brand strategy and visual identity toolkit, website design and editorial process embedded in the digital communication strategy; (iii) development of the methodology for consumer involvement and citizens awareness raising, based upon user-centred research and product design, training, gamification and the implementation through the use of Fab-lab network and living labs; (iv) establishment of circular value chains supported by the capacity-building programmes destined to all involved stakeholder groups; (v) Key Performance Indicators to monitor success of communication, dissemination and exploitation; (vi) Risk analysis and data management plan. Dissemination approach builds upon synergies with EU projects, clusters and the CCRI. **G_MEDIA,** a knowledge company, founded in 2012 and created to support partners in sustaining value through excellence in management and communication in the context of change and development, will be in charge of developing and continuously updating the **Dissemination & Communication activities.** The role of all partners in the dissemination activities is listed in Table 12.

Table 12 Specific partners activities for the Dissemination and Communication plan of the RECREATE results

PARTNER	DISSEMINATION & COMMUNICATION ACTIVITIES
G_MEDIA	Dissemination, Communication, Exploitation WP leader; responsible for the coordination of public awareness, community engagement, dissemination and communication strategy and implementation. G_MEDIA will drive the process of identifying the stakeholders to target, the modalities to reach and engage them and the other feature to ensure a targeted awareness raising and dissemination of the project results and active involvement of the stakeholders. A time-defined Dissemination Action Plan will be designed including the project web site design, implementation and curation. Two internal knowledge exchange meetings will be organized.
FRAUNHOFER	Communication and dissemination towards the wide network of connected stakeholders. Dissemination of scientific and technological project results through publication of scientific articles.
RESCOLL	Presentation of RECREATE on Rescoll's blog and social media. Participation to events and fairs related to composites (JEC world, Composites Meeting, ECCM). Exploitation of the project results on debondable adhesives (sales) and laser dismantling (services)
IRIS	IRIS as technology provider for smart recognition and monitoring system for fiber reinforced composites will support the dissemination and communication activities described below, by means of participating in relevant events or conferences, publications, and providing diffusion of the project and its results through its social media.
COBAT COMPOSITI	Dissemination towards the wide network of industries and companies connected, as a composite related waste manager. Communication to the public authorities and the legislation bodies by exploiting the available network of stakeholders at Italian and European level.
POLIMI	Dissemination and communication activities towards the scientific community in the material science and manufacturing domains within the participated scientific and industrial networks (Eucia, CIRP, IFAC), communication towards regional and national authorities and industrial facilitators, co-design and preparation of learning materials for supporting education of current and future generation of employees. Dissemination of scientific and technological project results through publication in open-access mode in high-impact scientific journals (see Table 15).
TAU	Dissemination and awareness-raising activities related to RECREATE results towards the whole composite industry in Finland.
HEAD Sport GmbH, REST UG, GEVEN SPA, EDAG, BENASEDO, GAIKER	Dissemination targeting specific customers, sectors and value-chains. Design and implementation of specific marketing strategies for the supporting the future market uptake and launch of the new demonstrated products.
INVENT	Design of specific dissemination, communication, and marketing campaign for supporting the future introduction of the innovative RECREATE green and digital technologies in the worldwide market, with specific interest on de-bonding technologies as novel design opportunities for customers will at designing and manufacturing new easy-to-disassemble sustainable products
AVK	Dissemination targeting potential future users of enabling demand-driven circular economy solutions at global level, for enabling industrial uptake and technology transfer to future business cases, according to the replication and multiplication. Dissemination towards the whole composite network, for animating replication both in mature and niche sectors.

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APRA Europe	Dissemination towards the automotive remanufacturing network for animating replication in different industrial business cases. APRA makes use of various well-established events, channels and partners such as Automechanika Frankfurt, APRA European Symposium, Social Media, own Newsletters.

With respect to Communication, the main channels that will be activated in the project are listed in Tables 13/14. Relevant ongoing/upcoming EU projects will be monitored, in link with the CCRI, for max cooperation opportunities.

Table 13 Stakeholder groups targeted by RECREATE communication activities.

Target groups	Aims (Benefits that RECREATE project will offer to them)	Main channels and methods
Recycling industry	Awareness, Involvement, Exploitation: Demonstration of economical viable mechanical and thermal recovery of composites.	Industry Associations, Exhibitions, Conferences.
Technology providers and design studios	Awareness, Involvement, Exploitation: Demonstrations of design solution supporting remanufacturing, online database of recovered fibers' properties	Industry Associations, Exhibitions.
European manufacturing industry	Awareness, Involvement, Exploitation: Strong industrial involvement towards developing new products from recovered and remanufactured glass and carbon fibers in high-value and high-strength applications. Inspiration and best practices for the development of new Circular Economy businesses.	Direct proactive communications. Industrial training. Demonstration workshops. Exhibitions (ReMaTec, JEC). Clusters. Vanguard ESM. Associations. Project web site. Social media. Education and Training.
European Technology Platforms	Awareness, Multiplication: Implementation of Strategic Research Roadmap's topics and successful demonstrators	Direct communication, input to strategy papers and research roadmaps.
Standardisation, certification bodies	Awareness, Involvement, Exploitation: New guidelines for sustainable circular economy services. Contribution to standards for EoL cloud-based platforms.	Direct Proactive communication. Participation to CENELEC working groups.
Professional Associations	Awareness, Understanding: Sharing guidelines with the associates. Involve associates to RECREATE training sessions.	Direct Proactive communication. Industrial training webinars and hands-on sessions.
Teachers and Students	Awareness, Understanding: Up-to-date information on CF and GF recyclate use in industry sectors. Attractive jobs in sustainable circular economy businesses.	Material to be used in education and vocational training.
Research community	Involvement, Exploitation: New methods for recovery and remanufacturing to be extended to other materials and technologies.	Scientific publications. Dissemination workshops.
Regional agencies and initiatives	Awareness, Involvement: Showcase of the implementation of Regional Smart Specialization RIS3 Strategies, involve in the preparation of the CEAP.	Direct communication, Demonstrators' Open-Door event.
Policy makers	Awareness: Recommendations for legislation guidelines supporting the business model of RECREATE also through the 'Innovation Deals'. Evidence of benefits of Circular Economy at large scale	Direct Communication; Clusters, White papers from associations, European Commission's events on Circular Economy.
NGO, Citizens	Awareness: Closing of material streams to the benefit of the environment	Project web site. Social media, Action under the EU Week for Waste Reduction Demonstrators', Open Door, Press release.

Table 14 Specific activities and methods for dissemination and communication in RECREATE.

Specific Channel	Scope and procedure for design, set-up and maintain the channel
The project image and identity	Image and visual identity will be documented in the stylebook. The image will include the logo, project specific design elements and templates for MS Office tools. Promotional material utilizing the visual identity will be generated (Prezi, SlideShare) and two Youtube videos will be produced to communicate the project's vision, objectives and results.
The project website	Will be produced by a professional marketing company to support the implementation of the project and it will be main interface for communication to the public. The website will host communication tools and a blog to facilitate interaction between partners and interested parties. The private section will facilitate the communication within the consortium.
Social media	Will be utilized by the consortium to communicate the project in professional social networks (e.g. LinkedIn). Official LinkedIn groups will be used to raise awareness of professional network groups.
The website blog	Will highlight outcomes, distil updates, presents the growth of the network and relevant news concerning exploitation opportunities and offering by the RECREATE companies.
The project factsheet	Will be prepared at start of the project and twice updated. The first version will focus on the project's objectives and the vision. The second version will highlight achieved results while the third and last version will demonstrate key results and uses cases.
Scientific publications	Will be increasingly submitted throughout the project to disseminate the results to the scientific community and experts in the industry. In the beginning of the project various conference participation will allow presentation of posters and publishing of abstracts. Thereafter, the accumulating results will be published in peer-review scientific journals.
Trade articles in professional magazines	Will be prepared to disseminate the results to experts and investment decision makers. Attendance on trade fairs and exhibitions are foreseen to present the project's results and to foster customer relationships and prepare for the commercial exploitation of results.
Press releases	The consortium will issue regular press releases at key milestones of the project by employing the main media communications (special websites, TV, newsletters, journals radio, etc.).

Regarding publications, the project will publish scientific and technical results in high impact open-access journals and relevant technology focused congresses. Table 15 identifies a list of target journals, conferences and fairs.

Table 15 Journals and Magazines, Conferences, Industrial Fairs and Exhibitions

JOURNALS	MAGAZINES
CIRP Annals - Manufacturing Technology, NDT & E International, Journal of Remanufacturing, Resource, Conservation and Recycling, Journal of Cleaner Production, Waste Management, Journal of Renewable Materials, Advanced Materials, ACS Sustainable Chemistry & Engineering, Composite Science and Technology.	APRA Newsletter, Lightweght Design newsletter, Manufacturing Global, Renewable Matter, Recycling Magazine, CompositesWorld, Composites Manufacturing Magazine, RemanWorld, JEC Magazine.
CONFERENCES	
CIRP Design Conference, CIRP General Assembly, CIRP Conference on Life-Cycle Engineering (LCE), JEC World, European Conference on Composite Materials (ECCM), International Conference on Remanufacturing (ICOR), Circular Economy European Summit (IEEE ICIT), IARC, SAE AeroTech Congress & Exhibition, Enterprise Integration, Interoperability and Networking, International Composites Congress (ICC), European Biomass Conference, Plastics Circular Economy Congress.	
INDUSTRIAL FAIRS AND EXHIBITIONS	
Automechanika Frankfurt; Rematec; EGG CONTROL; ECOMONDO; Plastics Recycling Show Europe; European Trade Fair & Forum for Composites, WindEnergy Hamburg, Husum Wind, JEC World Paris, Singapore Air show, International Paris Air Show Le Bourget, Luftfahrtmesse Farnborough, Dubai Air Show, Aircraft Interiors Expo, Mouldtec, EXIST, World Remanufacturing Summit, World Circular Economy Forum – WCEF.	

2.2.2.1 KPIs for the C&D&E activities

In order to support the ambitions of RECREATE, equally ambitious target KPIs have been defined for RECREATE. The achievement of these KPIs will be constantly monitored and reported in WP8.

Table 16 Key Performance Indicators for C: Communication, D: Dissemination and E: Exploitation activities.

ID	Indicator	Number of:	Year 1	Year 2	Year 3	Year 4
KPI1	C+D through the internet	RECREATE unique website visits	1500	2500	5000	10000
KPI2		Other websites linking to the RECREATE website	25	40	60	70
KPI3		Website updates	10	20	30	40
KPI4		Contributions to discussions in social media (Youtube, Linkedin, Twitter, Facebook)	20	35	45	45
KPI5		Uploaded presentation to SlideShare/Prezi	4	6	12	20
KPI6		Number of press release	5	10	15	20
KPI7	C+D through general media	Media coverage (print, radio, TV)	5	8	10	10
KPI8	C+D through presentations	Scientific/technical presentations	6	10	20	40
KPI9	D through scientific publications	Submitted publications in Journal and Conferences	2	6	10	15
KPI10	D activities promoting the results' exploitation	Events attended to present the demonstrators	3	10	18	22
KPI11		Industry participants in the exploitation workshop	N/A	N/A	60	120
KPI12		Requests from industry	15	30	50	100
KPI13		Showcased products in the RECREATE library	N/A	N/A	20	50
KPI14		N° of training webinar modules for industry and academic			8	15

2.2.1.2 Contributions to the EC policies in the 2020 Circular Economy Action Plan

According to the G7 Summit Declaration of June 2015: "The G7 Alliance on Resource Efficiency promotes Circular Economy, Remanufacturing and Recycling as strategic actions for limiting the consumption of natural resources and reducing waste". Moreover, at EU level, the "2020 Circular Economy Action Plan for a Cleaner and More Competitive Europe" sets policy "sustainability principles" among which (i) enabling remanufacturing and high-quality recycling and (ii) increasing recycled content in products, while ensuring their performance and safety, (iii) incentivising product-as-a-service or other models where producers keep the ownership of the product or the responsibility for its performance throughout its lifecycle; (iv) mobilising the potential of digitalisation of product information, including solutions such as digital passports, tagging and watermarks; (v) introducing a ban on the destruction of unsold durable goods. These are core pillars of the demand-driven circular approach of RECREATE. RECREATE will directly contribute to the implementation of these political orientations by proposing value-chain oriented solutions combining suitable sets of options for composite products post-use and re-use with a cross-sectorial approach tacking all the aforementioned sectors.

2.2.2.3 Contribution to relevant legislation

The RECREATE consortium conducted an in-depth analysis of Regional and European regulatory frameworks that may potentially create boundaries for the market exploitation of the RECREATE cross-sectorial circular economy value-chains. The output of this analysis is briefly summarized in Table 17 and specific actions described in T8.5.

Table 17 Relevant norms and legislation for the RECREATE cross-sectorial circular economy value-chains.

EU Norm and subject	Implications for RECREATE	Contribution of RECREATE (T7.9)
General Directives on waste		
List of Waste 2000/532/EC: Waste classification.	EU Waste list in relation to the Waste Framework Directive 75/442/EEC.	The most relevant six categories for FRP waste are: 12 01 05, 16 01 19, 17 02 03, 17 02 04* (M), 19 12 04, 20 01 39. RECREATE will support the potential identification/definition of new categories.
Directive on End of Life of Vehicles 2005/53/EC; EPR Legislation 2002/96/EC: Reduction of waste arising from ELVs.	Clear targets and timelines, independently on the materials adopted in cars. The directive involves four major stakeholders, the producer, the recycler, the last holder and the authorities.	Absence of minimal requirements for manufacturers involvement in EPR schemes. Requirements given only by total weight of recycled/re-used material. RECREATE will provide inputs for the extension of similar regulations for other product categories, such as wind turbines waste, and for clarifying the flow of responsibilities between value-chain stakeholders.
Directives on waste transportation		
Regulation 1013/2006: Waste shipment regulation.	Defines procedures and control regimes for the shipment of waste, depending on the origin, destination and route, the type of waste and treatment at its destination.	RECREATE will provide inputs for Regulation advancement through the potential adoption of the developed digital solutions for the full traceability of the composite waste, from its origin to the treatment facility.
Directives on waste treatment		
Directive on Waste 2008/98/EC: Waste management.	Sets the basic criteria to characterize, to declare and to re-commercialize ex-waste to a new "product".	Poor consideration of remanufacturing. Absence of standards for the quality of secondary raw materials. RECREATE will provide inputs for introducing standards for traceability of materials and for standardized certification of high-quality materials and components.
Eco-design Directives		
Eco-design Directive 2009/125/EC: Sustainable product design	Attention to the product design phase and establishes a framework to set mandatory ecological requirements for energy-using and energy related products sold in all 28 Member States.	Requirements and labels are mainly related to the use-phase energy and not to manufacturing operations. Eco-design is not considering the product design for disassembly and re-use. RECREATE will provide suggestion to improve requirements and labels with material efficiency indicators such as reusability, reparability and recyclability. Focus on composites products design for remanufacturing and on co-designed products embedding re-usable materials.

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2.2.3 Management of research data, knowledge and Intellectual Property Rights (IPR)

The management of research data, knowledge and IPR is an integral part of the project's innovation management. RECREATE will yield extensive research data sets (Table 18). The project's Quality Assurance Plan (QAP), including the "Good research practice", will ensure in-detailed documentation of research data (T9.1). The consortium is committed to open access principle and supports sharing of results where appropriate. Results will only be published with IPR protection clearance. The consortium will avoid disclosure of business sensitive data by non-disclosure or aggregation for numeric data. Owners of the data can decide on the openness and usage policies for their own data. Data ownership, especially of data generated by and obtained from industrial research and prototype testing, will be identified in the CA.

Table 18 Sub-set of data generated in the project and to be made available to external parties.

Data	Origin	Format	External Sharing
Raw data from experiments	Validation and verification tasks in WP2,3,4,5,6.	Data sets, excel files, text files, html files.	Only upon request and respecting IPR agreement.
Generalised, agglomerated data and metadata	Validation and verification tasks in WP2,3,4,5,6.	Graphs showing qualitative materials behavior.	Via open access publications. Metadata repositories
Product Eco-Design guidelines	Deliverable 2.7	Instructions, pictures, technical details of co-designed products.	Public access will be given. Download from the web-site.
Idea Manager files	Deliverable 5.5		
Virtual and Product Library files	Deliverable 5.5	3D animation and rendering, co-designed products (text).	Public access will be given to the Library to stakeholders.

2.3 Summary

SPECIFIC NEEDS

Composite materials still lack of a systematic process-chain for ensuring recovering and re-use and of a stable value-chain. Circular economy potentials are still weakly explored at industry level and they frequently end-up in landfills.

There is a lack of economical and environmental sustainable technologies and circular business cases for high-value re-use of composite materials meeting the demanded specifications in cross-sectorial applications.

Integration of value-chain stakeholders and scouting of new market application potentials are still unexplored due to challenges in establishing cross-sectorial interactions.

EXPECTED RESULTS

Demonstration and validation of the effectiveness of the demand-driven circular economy concept in overcoming the current challenges and fulfilling the needs.

Successful validation and demonstration in 4 prototypes of new products embedding recycled materials and in 5 prototypes of new remanufactured components.

New methods and digital tools for process chain control and adaptation.

New technologies supporting re-use of components, including LIBS for composite identification and sorting, laser-based components disassembly, thermally-assisted reforming, remanufacturing and repair.
New demand-driven technologies for recycling of composites fibers and resins including electrofragmentation and catalyst-assisted solvolysis.
New sustainable and easy-to-disassemble composite structures amd reversible composite materials.

DISSEMINATION EXPLOITATION AND CCOMMUNICATION MEASURES

A list of 5 types of exploitable results have been populated including Added-value, customized products re-using fibers and resins, FRP components modular design, reman., repair, reforming, and re-use, new sustainable-by-design composite materials with enhanced functionalities, Circular Economy digital platform for supporting the adoption of products re-using materials and components, Digitalisation of knowledge about new materials, products and innovative processes for composite re-use for training tools and gamification. These include harware and digital enabling technologies and at least 9 new products. The strategies for exploitation fo these results is reported and will be further refined through 4 targeted workshops.

Dissemination towards the composite industry in different sectors by exploiting the presence of important associations such as AVK and APRA Europe. Scientific publication with the results of the large-scale demonstration.

Scientific publications in the most relevant journals in the field of composite materials, automotive, energy, manufacturing, recycling, remanufacturing and circular business models.

Co-design of training materials and gamification program by developing e-learning contents and digital games targeted to a wide set of groups.

Communication towards citizens: participation to the Milan Design week to showcase the sustainable products.

TARGET GROUPS

European SME, both end-users and technology providers.

Composite component manufacturers. Composite material suppliers. Resin suppliers.

SME in the B2C context, designers, cross-sectorial business facilitators (Clusters)

European citizens at large, manufacturing companies in strategic EU sectors.

Public and legislation bodies.

Scale:

Through the dissemination, exploitation and communication activities of RECREATE at least 30 new re-use business cases will be identified during RECREATE that will be further advanced after the project through local clusters, such as AFIL (Lombardy), Merge (Saxony), Polymeris (France).

At least 5 large scale dissemination and demonstration events targeting industries not represented by RECREATE partners and designers (e.e. Milan Design Week).

Replication and multiplication by exploiting the Vanguard Initiative De-and Remanufacturing network including 8 regional hubs for Circular Economy network,150 EU stakeholders, more than 100 de-risking project expected in the next 5 years, potential turnover of about 500MEuro generated for European SMEs using the hubs for exploring their transformation path towards circular economy.

OUTCOMES

9 use-cases demonstrating feasible new cross-sectorial business cases, with positive economic assessement and LCA results.

Massive reduction of composite waste going to landfill and downgraded applications.

Increased uptake of novel composite materials including reversible resins, reversible joining adhesives, new modular design of automotive structures for future vehicles in the e-mobility.

6 new de-and remanufacturing technologies to be sold by SMEs to the world wide market .

Replicable solutions for more circular composite products, through the digital plaform and the exploitation of the VI De-and remanufacturing circular hubs network.

Increased awareness and trust on circular products through the ambitious Dissemination & Communication & Exploitation plan.

IMPACTS

Economic: Demonstration of new cross-sectorial business cases for re-using glass and carbon fibers and resins in high-added value products.

Size: 0.203 B€ cost savings in the considered, conservative business scenario by 2030, related to reduced cost of fibers (glass: -40%, carbon: -70% with respect to virgin) with in specification characteristics, and remanufacturing and re-use of composite structures.

Societal: More knowledge intensive jobs in the targeted industries (+4149 by 2030) due to new circular economy operations.

Environmental: Avoid landfilling and downcycling of composite materials. In the considered scenario, 51.3 kTons of waste composites per year, 2.64 TWh/year of energy, and 1555 kTons CO_2/year of emissions can be saved.

Scientific: New breakthrough sand multi-disciplinary scientific results on a wide range of topics including Digital tools and Green Technologies for a new vision of demand-driven Circular Economy.

3. Quality and efficiency of the implementation

3.1 Work plan and resources

In line with the scientific and technological objectives of RECREATE, the work in this **48-month** project has been structured in 9 work packages (WP, Fig.16), each sub-divided into a number of dedicated tasks covering the activities planned for the project, as listed in detail in **Table 4**. The project coordinator will be Prof. G. Griffini, associate professor in Materials Science and Technology at POLIMI. Each WP will be led and managed by a project partner who is expert in the relevant field, as described in detail in Section 3.2. This WP structure aims to set the basis for a rational and seamless execution of project activities and successfully achieve the foreseen objectives, implemented in a reasonable and cost-effective manner. The scientific, technological, techno-economical, environmental and educational challenges addressed in RECREATE are in line with the HORIZON-CL4-2021-RESILIENCE-01-01 topic. Active collaboration between all 21 partners will generate new scientific knowledge, develop novel, sustainable and economically profitable technologies, implement novel approaches to learning and education in the field of composite circularity.

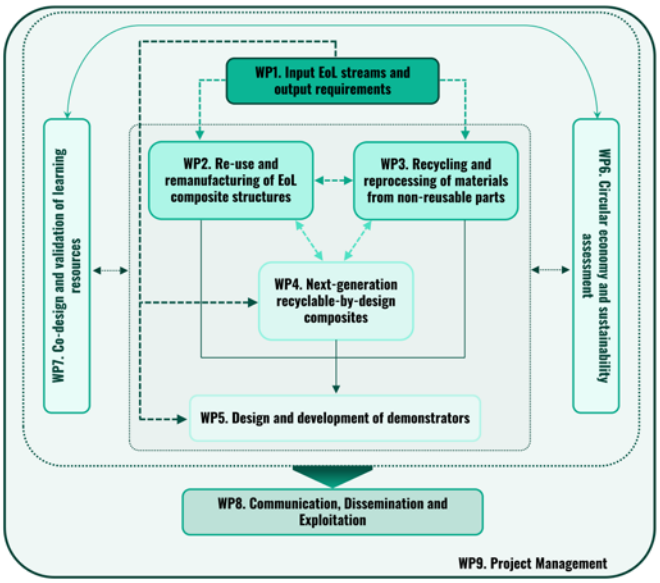

Fig. 16 WP interconnections and workflow.